Measuring integrated delivery of essential public health functions

A review of health systems and health security monitoring tools

World Health Organization

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Abbreviations

EPHF	Essential public health function
HHFA	Harmonized health facility assessment
IDSR	Integrated Disease Surveillance and Response
IHR	International Health Regulations
PAHO	Pan American Health Organization
SPAR	States Parties self-assessment annual reporting
WHO	World Health Organization

How to use this report

This technical report is divided into two main parts.

Part A (subsections 1.1–4.8) introduces the technical report and provides the technical foundation and evidence base. Most of Part A (sections 2–4) is dedicated to reporting the evidence generated to further inform country-level policies and actions for ensuring integrated measurement of the essential public health functions (EPHFs) as part of health system strengthening and resilience building. Part A is therefore useful for better understanding the strengths and limitations of current approaches to monitoring public health functions (particularly health promotion, disease prevention, health protection, emergency management and community engagement aspects) with reference to major examples of measurement tools widely used in routine and emergency contexts.

Part B (subsections 5.1–5.5) provides guidance for country-level actions, building on available evidence as well as expert and stakeholder experiences in measuring and monitoring EPHFs. Country examples and tools, including an adaptable checklist, are also provided to support countries in their efforts to ensure more integrated and comprehensive measurement of EPHFs. Part B is therefore useful for identifying and applying practical steps, examples and tools for country-level decision-making and actions.

Annex 1 provides an expanded cross map between the key monitoring tools – reviewed when developing this technical report – and the 12 EPHFs which can inform further analysis, given that this report focuses on 5 out of the 12 EPHFs.

Annex 2 is a technical brief developed based on this report to support communication and advocacy targeting policy and decision makers in countries and at regional and global levels.

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Part A

1. Introduction

1.1 Background

Countries are faced with complex public health landscapes with concurrent multifaceted challenges, including emerging and re-emerging infectious disease outbreaks, multimorbidity of noncommunicable diseases and mental health challenges, migration and displacement, and profound impacts from climate variability and change. This situation requires national health systems and allied sectors to develop and maintain cross-cutting, comprehensive, flexible and foundational public health capacity to manage these chronic and acute stressors with clear and shared accountability. The essential public health functions (EPHFs) are a set of fundamental, interconnected and interdependent activities required to ensure effective public health action to prevent disease, promote and protect health and well-being, and address broad health determinants emanating from environmental, social, economic, cultural, lifestyle and biopsychosocial drivers. The World Health Organization (WHO) has defined a unified list of EPHFs consisting of four main service-oriented and eight enabling EPHFs. World Health Assembly resolution WHA69.1, 2016, recognizes EPHFs as the most cost-effective, comprehensive and sustainable means of enhancing population health and reducing disease burdens *(1)*, which is also a necessity for building health system resilience in support of the complementary goals of universal health coverage, health security and healthier populations. Alongside this resolution, the Sixty-ninth World Health Assembly adopted a framework on integrated, people-centred health services, recognizing the importance of improved access to care, improved health and clinical outcomes, better health literacy and self-care, increased satisfaction with care, improved job satisfaction for health workers, improved efficiency of services, and reduced overall costs *(2)* – all of which are associated with the EPHF approach.

The EPHF framing enables strong connections to be built within and beyond the health sector, where responsibility for public health rests, as it encompasses service-level functions as well as functions that enable public health services (including promotive, preventive, protective and emergency management functions) *(3)*. At the national level, coordinated planning of, investment in, and institutional arrangements for EPHFs could reduce the current prevailing fragmentation and improve the self-reliance and resilience of health systems while leveraging multisectoral efforts for health. Health and allied sectors can anchor their planning and investment efforts within the foundations of society and the health system, addressing common requirements in response to diversified public health demands. On the other hand, the subnational level is the first point of contact for emerging threats and bears the initial impact of emergencies. In this context, EPHFs enable consideration of comprehensive and integrated provision of public health services within front-line subnational structures (for example, primary care, communities, district health management and local health administration), including those outside the health sector, so that they are empowered to better serve the local population routinely and manage public health emergencies. The flexibility and integrated nature of the EPHFs also allows countries to adapt them to their context; for example, in Canada the EPHFs consist of six functions, which include the four main service-oriented EPHFs (health promotion, disease prevention, health protection and emergency management) in WHO's global list of EPHFs *(4)*.

Despite global recognition and commitment, country and global experiences with public health challenges have demonstrated that public health practice, operation and services are often absent or not well defined between jurisdictions and across national, subnational and primary care and community levels (5). To guide the global discussion and national implementation, WHO has developed a range of technical resources and flexible tools on EPHFs, which can support planning for the comprehensive operationalization of public health in countries. To move forward in country application, monitoring and evaluation of comprehensive and integrated delivery of EPHFs and public health services need to be understood and strengthened. Detailing how health system monitoring approaches are capturing EPHFs can enable better technical understanding and can help guide shared accountability for public health. This builds on the work of bundling public health service and system enablers, supporting and guiding countries to undertake benchmarking, connecting dots across sectors and allowing monitoring with visible oversight for public health delivery. The interconnected nature of measurement of EPHFs provides a sustainable route to monitor progress on integration and people-centredness, as highlighted in the Sixty-ninth World Health Assembly. This is also aligned with the WHO Fourteenth General Programme of Work output on strengthening institutional capacity for EPHFs to improve health systems resilience (6).

1.2. Aim and Scope

The aim of this technical report is to determine the adequacy of key health system and health security monitoring tools for integrated measurement of EPHFs and to provide guidance for improvement, with a focus on integration. While considering the full range of EPHFs, this technical guidance focuses on the service-oriented EPHFs (health promotion, disease prevention, health protection and public health emergency management) and community engagement that is culturally sensitive, considering the centrality of people to the delivery and measurement of public health services in all contexts.

This review examines selected health system and health security measurement tools[1] used in routine, voluntary and emergency contexts for their current scope to support the assessment of public health functions (community engagement and service-oriented EPHFs), including their integrated delivery. It considers measurement tools within the health sector only and their interaction and overlap with other allied sectors with potential roles in public health services and functions.

1.3. Target audience

This technical report is primarily intended for policy-makers, senior public health managers and emergency managers at national and subnational levels who are linked to processes of planning, implementation, monitoring and evaluation of public health policies, programmes or projects, including those working in ministries of health, national public health institutes, and senior health service and sector management, in addition to those working in ministries that influence population health. It can also be used by those in the humanitarian and development sectors that support countries in strengthening health systems and public health capacities, including relevant entities of the United Nations community, networks of public health institutes and professionals, academic schools, research councils, foreign aid agencies and donor organizations.

1.4. Key concepts and technical foundation

1.4.1 Integrated delivery of public health functions

The added value of the EPHF framing is largely derived from its positioning as an integrated and comprehensive approach to meeting population

[1] This denotes the monitoring tools available for measurement of health system functions, performance and service availability related to universal health coverage, health security and public health.

health needs. As EPHFs are widely dispersed within and beyond the health sector, it is crucial to pursue integration-enabling multisectoral collaboration and accountability. Integrated delivery of public health functions entails a complementary combination of functions with coordinated and collaborative participation of all stakeholders, as needed to address public health problems in a holistic manner. It also requires incorporating and mainstreaming the EPHFs within existing public health systems (across health and allied sectors) to sustainably strengthen and build resilience in the system. For example, to tackle road traffic accidents as a public health issue, ministries of transport, health, education, security agencies, and urban planning, as well as civil society and the media, will need to work together to implement a combination of EPHFs ranging from public health surveillance and governance to promotive, preventive and protective services with community engagement.

While there is global acknowledgement that integration is essential to the successful application and measurement of the EPHFs, current evidence points to fragmented and siloed approaches in tackling public health issues. As an example, incorporation of health promotion – with its focus on culturally appropriate community engagement and participation *(7)* – in emergency preparedness and response can address many of the risk drivers and mitigate the impacts of a wide variety of public health threats. Despite this, health promotion and community engagement approaches were infrequently embedded within emergency preparedness and response activities, and where mechanisms did exist, they were often not leveraged early in the COVID-19 pandemic *(8)*. Further recent public health challenges, including the spread of mpox into new contexts, demonstrate the benefit of communities being central actors in managing the public health emergencies they are facing, enabling a more effective and sustainable response *(9)*. The unified list of public health functions published by WHO as a definitive set of 12 EPHFs gives prominence to community engagement alongside emergency management and health promotion, among other EPHFs, and reiterates the need for them to be proportionately considered and delivered together and in synergy, reflecting population health needs (Table 1) *(5, 10)*.

Table 1. Essential public health functions by service-oriented and enabling-oriented functions

Public health service-orientated functions	Public Health Emergency Management
	Health Protection
	Disease Prevention and early detection
	Health promotion
System inputs and enabling-orientated functions	Community Engagement and social participation
	Public health workforce development
	Public health stewardship
	Multisectoral planning, financing and management for public health
	Health service quality and equity
	Public health research, evaluation and knowledge
	Access to and utilization of health products, supplies, equipment and technologies
	Public health surveillance and monitoring

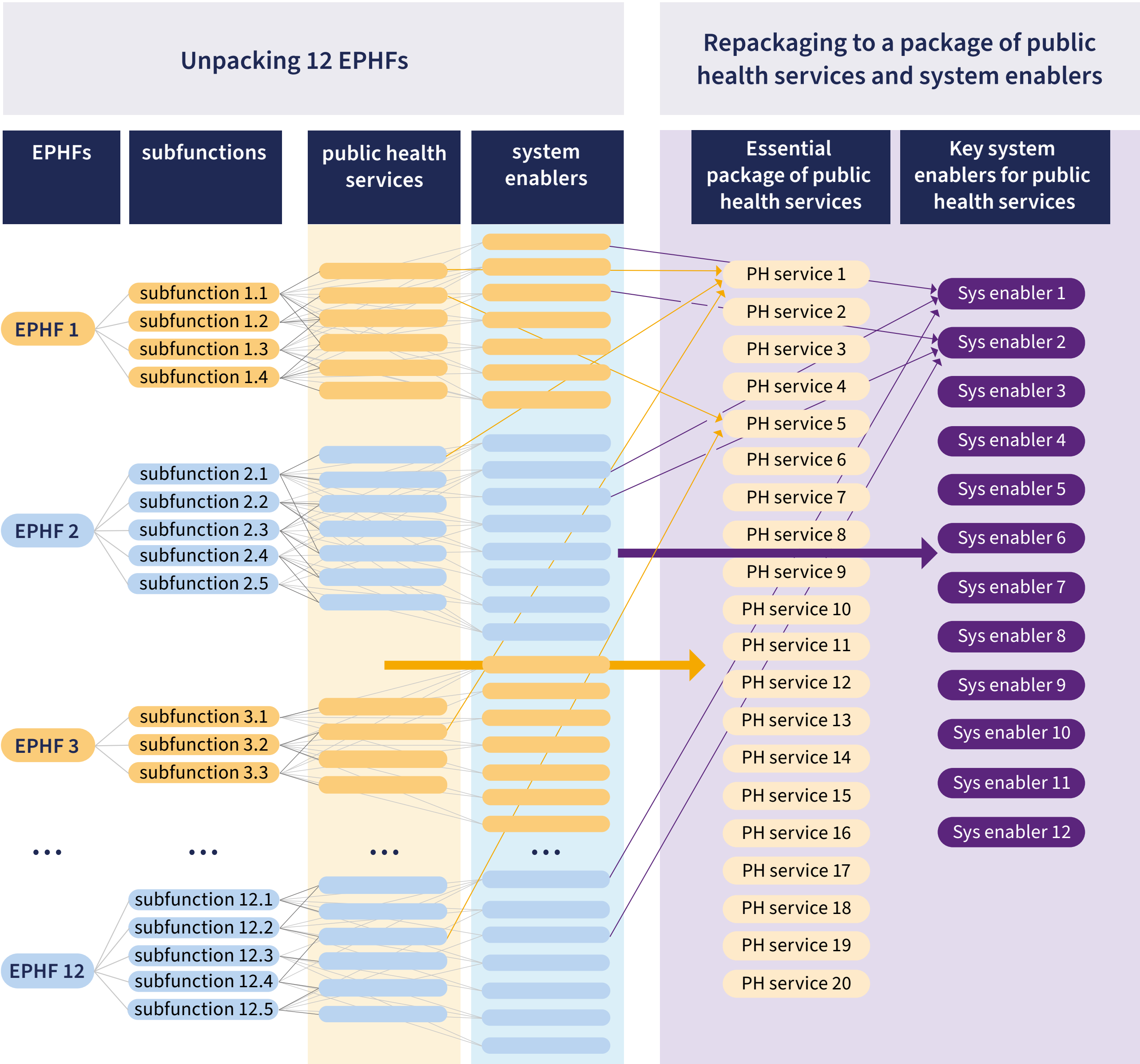
The list of EPHFs also identifies and provides emphasis on four service delivery-focused public health functions along with eight enabling and cross-cutting functions. Community engagement and social participation is identified as an enabling function for all four service-oriented functions and a central consideration and input for the other seven enabling and cross-cutting functions. In order to measure such integrated delivery of the EPHFs, a focus is needed on the interconnections and interdependencies between these areas, which mutually reinforce and support each other. To support the operationalization of integrated delivery, the 12 EPHFs and their constituent services and system enablers were further delineated into subfunctions using standardized definitions (Figure 1). Each of these elements were then unpacked and reorganized into a more streamlined and operational list consisting of 20 public health services and 12 system enablers required for delivery of these services (Table 2). Box 1 outlines definitions of public health services and system enablers.

Box 1. Definitions of public health services and system enablers for public health services

Public health services are defined as actions with a primary focus on improving population-level health outcomes, including promoting health equity, while reducing risks and promoting health at the individual level. Public health services reflect a wide range of actions (for example, regulatory instruments, such as policy and legislation development and implementation; fiscal instruments, such as fiscal measures to promote healthier choices; clinical services; social measures; information and education instruments, such as knowledge, awareness, attitude and behaviour change, communication and advocacy; and environment modification) that seek to positively impact the broader determinants of health and wider issues in the promotion and protection of health, including those across various sectors, such as health, agriculture, environment, commercial services, education, transport and housing.

Public health system enablers are the public health infrastructures, capacities, institutional arrangements and processes that are required to ensure the comprehensive and integrated delivery of public health services.

Figure 1. Schematics of EPHF and subfunction unpacking and organization into package of public health services and system enablers



1.4.2 Health system measurement

Measurement of public health functions must go together with health system performance assessment and monitoring. A health system cannot perform well without public health functions and services, as the health system would not be able to anticipate, prepare for, manage and recover from shocks and stressors that require routine public health information, planning and action. Moreover, many aspects of public health governance and delivery, as well as the determinants of health, lie beyond the health system, requiring close collaboration and joint working across sectors. Figure 2 provides an illustration of the dispersed and multisectoral distribution of public health governance and implementation structures in countries (*11*).

Figure 2. Illustration of the dispersed nature of public health functions and services, including their measurement, within and across health and allied structures in countries

Measurement and reporting drive priority setting, particularly in areas such as public health, where resources are limited and – simply put – what gets measured gets more attention and is more likely to get done. There are many examples of how defining and incorporating indicators for measurement has driven decision-making in emergency and routine settings – for example, risk communication has received increased prominence in public health emergencies, largely because of being measured in reporting tools such as the International Health Regulations (IHR) *(2005)* States Parties self-assessment annual reporting (SPAR) and such assessments as Joint External Evaluation. Although there was agreement that supporting risk communication capacities was needed, it was often not prioritized until it was routinely measured and reported on.

Currently, there is no one single process to systematically monitor all EPHFs across routine, ad hoc and emergency-focused health information systems. Existing health systems and health security assessment tools should be well placed to facilitate and enable such monitoring. Though designed for specific purposes, they could also apply a more comprehensive public health lens to capture information useful for measuring public health functions as applicable to the scope of the tool. This report will consider how essential public health functions are being measured through existing health system monitoring and performance tools and identify areas where further specific measurement may be required.

2. Methodology

2.1 Health systems assessment tool selection

In line with the aim, relevant health systems and health security measurement tools were identified and assessed. Selection criteria for inclusion of tools were as follows:

- publicly available and freely accessible measurement tools;
- tools in regular usage by WHO Member States across a range of contexts;
- tools which contained indicators that covered health system performance or health security measures.

Tools were excluded if they were limited to indicators only measuring clinical outcomes or were focused on future planning only rather than assessment of existing systems and services for health.

Health system measurement tools were selected across three major categories:

- routine health systems and health security measurement tools;
- voluntary health systems and health security measurement tools;
- emergency health systems and health security measurement tools.

Tools included for review under each category were selected due to their broad relevance and implementation in countries. The core components of each tool were gathered and mapped to the 12 essential public health functions (Table 1). All tools were then further mapped to the 47 subfunctions that disaggregate the essential public health functions into their component elements *(3)* (see Annex 1). However, this technical report mainly focused on the results drawn from the four service-oriented EPHFs and community engagement. The choice of tools represents the consideration of EPHFs from routine to ad hoc monitoring and those applied during emergencies (such as COVID-19) and humanitarian contexts.

2.2 Approach to mapping existing tools with EPHFs

2.2.1 Review of indicators

Each indicator in existing tools was reviewed against the 47 subfunctions and assessed to see if it provided either indicative or partial levels of information that contributed to measuring the implementation of the corresponding public health capacity (Table 2). Determination of the "partial" level of public health capacity measurement of each subfunction represents a range of coverage for which further disaggregation was not in the scope of this review. The indicators that did not fall into any of these categories were left out as they either provided information limited to the results or outcomes, for example, mortality or morbidity indicators (not implementation) of the public health subfunction being examined, or were not at all focused on the public health subfunction.

Table 2. Categorization of public health capacity measurement across EPHF subfunctions

Level of public health capacity measurement	Description	Explanation	Example
Indicative	Indicator or indicator set provides adequate information to determine capacity to deliver a given public health subfunction	Existing indicator or indicator set may not match the exact language of EPHFs and may not fully reflect the integrated or cross-cutting ethos of the EPHF package but provides sufficient information for a policymaker to determine the capacity for delivery of this subfunction	Indicator on existing surveillance system captures core public health data
Partial	Indicator or indicator set provides some information that is of use to determine capacity to deliver a given public health subfunction	Information provided by the indicator or indicator set needs to be supplemented by other indicators or data to enable a policymaker to determine availability of capacity to deliver the subfunction	Indicator captures single dimension of public health workforce capacity
Absent	*The indicator does not provide information to determine capacity for delivering a given subfunction*	*Information provided by the indicator or indicator set is not aligned with the public health subfunction*	*Indicator does not capture or is unclear regarding any aspect of health promotion*

2.2.2 Integrated measures

An additional note was made where an indicator represented an integrated measure of different public health capacities or functions (for example, community engagement embedded in emergency preparedness and response). This included instances where one EPHF was measured within the primary context of a different function, or the interlinkages between different aspects of EPHFs were represented in a particular measurement (for example, ensuring quality health service continuity in emergency preparedness and response), or elements of different EPHFs were measured together in combination. Integrated measures are of particular importance as they provide evidence of how public health functions bridge areas of work, priorities and investments, promoting systems thinking and coordination as necessities for building strength and resilience into public health systems.

For measurement purposes, each public health function was considered in turn, but in practice, all essential public health functions should be interdependent, and delivery can only be effective where they are fully integrated and aligned. Significant overlap also exists between public health areas; for example, timely public health emergency management strongly relies on effective health protection and disease prevention functions.

2.3 Analytical approach

Within the scope of this document, analysis of the adequacy of existing health systems and health security measurement tools was focused on the four main service delivery-oriented public health functions. These include health protection, disease prevention, health promotion and public health emergency management, with the community engagement function also considered, given the centrality of community-centred approaches to integrated delivery of the EPHFs. This approach supports a targeted focus on public health service orientation, as the measurement tools are oriented towards health systems and services rather than focused on enabling public health functions. A note was made where the assessment considered the integration and interdependencies across these essential public health functions.

The subfunctions of each main public health function focused on in this review are listed in Table 3, with an integrated approach adopted to the review of existing health and health security measurement tools. This is particularly important in the context of community engagement, which should be considered within emergency management, health promotion and other service-oriented public health functions.

Two areas of thematic guiding questions were adopted to support the analysis of selected tools across service delivery-oriented public health functions and community engagement:

- Question area 1: What are the stated purpose and measurement intention of existing health system monitoring tools in relation to EPHFs?
  - What are the objectives of the tool, and do they include public health capacity measurement?
  - What are the overall structures and components of the tool?
  - What is the main target context for the use of monitoring and assessment tools?
- Question area 2: How do components of health system monitoring tools compare to EPHFs, and where are strengths and gaps noted?
  - How do tool indicators and EPHF subfunctions compare in a systematic comparison?
  - What level of public health capacity has been measured (as per categorization as indicative or partial measure of public health capacity in Table 2)?
  - What elements of the monitoring tool are integrated across EPHFs, including community engagement, health promotion and public health emergency management?
  - What strengths and gaps are noted in existing health system monitoring tools for measuring the integrated delivery of EPHFs?
  - What needs to be addressed beyond the existing tool to provide an integrated measure of EPHFs at the country level?

For visual representation, a high-level weighted representation of the relative contribution from each tool or set of tools towards the measurement of the specified functions was developed and represented using spider diagrams.

Additionally, an expert consultation workshop was held on 30 September and 1 October 2025, with global experts from the International Association of National Public Health Institutes, selected national public health institutes and WHO for a joint review of draft findings. Expert feedback received during this workshop was collated and utilized to finalize the report. Participating experts were affiliated to governmental entities (e.g., ministries of health, national public health institutions), or international nongovernmental organizations in official relations with WHO or with a Memorandum of Understanding with WHO in this area. There were no potential or reasonably perceived conflict of interest related to the subjects discussed at the meeting or covered by the report.

Table 3. Exemplar public health functions and subfunctions included in review and used as the focus of this document

Public health emergency management (EPHF 2 in accordance with the list)
Subfunction 2.1: Monitoring and analysing available public health information to identify and anticipate potential and priority public health risks, including public health emergency scenarios
Subfunction 2.2: Planning and developing capacity for public health emergency preparedness and response as part of routine health system functioning in collaboration with other sectors, including development of a national health emergency response operations plan
Subfunction 2.3: Carrying out and coordinating effective and timely public health emergency response activities while supporting the continuity of essential functions and services
Subfunction 2.4: Planning and implementing recovery from public health emergencies with an integrated health system strengthening approach
Subfunction 2.5: Engaging with affected communities and stakeholders in the public and private sectors and health and allied sectors as part of whole-of-government and whole-of-society approaches to public health emergency management
Health protection (EPHF 5 in accordance with the list)
Subfunction 5.1: Developing, implementing, monitoring and evaluating regulatory and enforcement frameworks, including compliance with international legislation, and mechanisms for the protection of specified populations (for example, workers, patients, consumers) and the general public from health hazards
Subfunction 5.2: Conducting risk assessments, risk communication and other risk management actions needed for all manner of health hazards
Subfunction 5.3: Monitoring, preventing, mitigating and controlling confirmed and potential health hazards
Disease prevention and early detection (EPHF 6 in accordance with the list)
Subfunction 6.1: Designing, implementing, monitoring and evaluating interventions, programmes, services and platforms for primary, secondary and tertiary prevention, including consideration of equity
Subfunction 6.2: Integrating consideration of prevention and early detection into service delivery platform design or redesign
Subfunction 6.3: Working with partners to support the development, implementation, and monitoring of legislation, policies and programme activities aimed at reducing exposure to risk factors and promoting factors that prevent disease
Health promotion (EPHF 7 in accordance with the list)
Subfunction 7.1: Designing, implementing and evaluating specific interventions or programmes to promote health, including changes in behaviour, lifestyle, practices, and the environmental and social conditions that promote health and reduce health inequities
Subfunction 7.2: Taking and supporting action, with partners, to address wider determinants of both communicable and noncommunicable diseases through a whole-of-government, whole of-society approach, including increasing individual and community participation in health-impacting decisions
Subfunction 7.3: Advocating, developing and monitoring legislation and policies aimed at promoting health and healthy behaviours and reducing inequities
Subfunction 7.4: Undertaking evidence-based advocacy and health communication to promote healthy behaviours and socioecological environments and build community trust
Community engagement and social participation (EPHF 8 in accordance with the list)
Subfunction 8.1: Promoting participatory decision-making, planning, research priority setting and science advisory processes for health and the promotion of societal change that enhances, promotes and protects health and well-being
Subfunction 8.2: Building community capacity for participating in public health planning, interventions, services, and preparedness and response measures
Subfunction 8.3: Monitoring and evaluation of community engagement in public health planning, interventions, services, and preparedness and response measures to promote equity and inclusion
Subfunction 8.4: Mobilizing and collaborating with communities and civil society groups in health services, interventions and programmes as part of a whole-of-society approach
Subfunction 8.5: Engaging communities in health preparedness, readiness, response and recovery

HHFA = Harmonized Health Facility Assessment; SPRP M&E = COVID-19 Strategic Preparedness and Response Plan: Global Monitoring and Evaluation Framework

3. Results

Findings are presented across existing health systems and health security measurement tools, focused across three areas:

- review of the stated purpose and measurement intention of existing tools;
- comparison of these existing health systems and health security measurement tools and their capacity to provide information on measurement of EPHFs, focused on service-oriented functions and community engagement;
- reporting on examples in existing tools of integrated measurement of EPHFs.

3.1 Purpose and intention of existing health systems and health security assessment tools

The health systems and health security measurement tools included in this review have been designed and iterated based on a range of contexts across stable and emergency settings and to meet specified objectives, which are reviewed here. The routine health system measurement tools included provide data continuously through health information systems (for example, DHIS2) embedded in the regular functioning of the health and surveillance systems (such as the Integrated Disease Surveillance and Response strategy) or on an annual basis, as specified in international statutes (as in the case of SPAR). A voluntary health system measurement tool – Harmonized Health Facility Assessment – provides additional metrics when countries choose to complete comprehensive health facility surveys. The emergency health system assessment tools included cover humanitarian contexts (for example, the Sphere Handbook) and the COVID-19 pandemic (pulse survey, Strategic Preparedness and Response Plan) as examples. Other measures, including the Health System Performance Assessment Framework for Universal Health Coverage and Health Systems in Transition reviews, provide a conceptual aid to orient analysis of information collected through health system assessment exercises or focus on informing the planning of health systems so they are not included as regularly used tools embedded in countries *(12)*. A number of stand-alone tools have been developed specifically to measure EPHFs (with indicators covering health system performance and health security measures). These EPHF assessment tools were not included in this review, as they are yet to be routinely or widely used in countries *(13, 14)*.

Although the included tools were not designed originally for the integrated measurement of EPHFs, the following assessment provides evidence of where existing data and methods are adequate to determine public health capacities in the contexts they are already used. Understanding the original intention and purpose of each tool in relation to the EPHFs supports understanding the value they bring to EPHF measurement.

DHIS2. The routine health system measurement tool DHIS2 (formerly known as District Health Information System or Software) is an open-source platform for the collection, reporting, analysis and dissemination of aggregate and individual-level health data, commonly used as a national-scale health management information system. The integrated health services analysis toolkit was used for analysis as it is the recommended indicator set for health system measurement using the DHIS2 platform *(15)*. As an integrated national health management information system, DHIS2 has a variety of use cases, with public health surveillance functions increasingly embedded into national DHIS2 platforms *(16)*. The integrated health services analysis toolkit has a range of indicators across three groups:

- Group 1 indicators - Health status and epidemiological profile:
  - mortality (institutional)
  - morbidity (inpatient and outpatient)
- Group 2 indicators - Health service performance:
  - utilization and access
  - service outputs, coverage and quality
- Group 3 indicators - Health service resources:
  - availability, distribution and efficiency of resources required by health facilities: infrastructure, health workforce, medicines and medical products, and financial resources

Integrated Disease Surveillance and Response (IDSR). IDSR is a strategy for implementing comprehensive public health surveillance and response systems for priority diseases, conditions and events at all levels of health systems *(17)*. It includes indicator-based surveillance and event-based surveillance approaches to early detection of priority diseases, conditions and events. As such, public health emergency management and health protection are central to its objectives. Health promotion, disease prevention and community engagement in contexts beyond emergency management are not clearly specified in the primary objectives of IDSR.

States Parties self-assessment annual reporting. The health security-focused tool IHR SPAR applies 35 indicators for the 15 IHR capacities needed to detect, assess, notify, report and respond to public health risks and acute events of domestic and international concern *(18, 19)*. The 15 capacities are as follows:

- C1. Policy, legal and normative instruments to implement IHR
- C2. IHR coordination and National IHR Focal Point functions and advocacy
- C3. Financing
- C4. Laboratory
- C5. Surveillance
- C6. Human resources
- C7. Health emergency management
- C8. Health services provision
- C9. Infection prevention and control
- C10. Risk communication and community engagement
- C11. Points of entry and border health
- C12. Zoonotic diseases
- C13. Food safety
- C14. Chemical events
- C15. Radiation emergencies

Harmonized Health Facility Assessment. This builds upon the Service Availability and Readiness Assessment and other assessment packages to provide a comprehensive health facility survey that assesses the availability of health facility services and the capacity to provide these services at the required standards of quality *(20)*. The Harmonized Health Facility Assessment includes four modules focused at the facility level, including service availability, service readiness, quality and safety of care, and management and finance. The stated objectives focus on health facilities and are not explicit in their consideration of public health services.

Pulse survey on continuity of essential health services during the COVID-19 pandemic. The pulse survey was conducted over four rounds and assessed the impact of the COVID-19 pandemic on essential health services *(21)*. It provided a rapid snapshot of changes and challenges in service delivery and utilization during the COVID-19 pandemic. It was also used to inform countries to support policy and planning dialogue on critical bottlenecks and guide mitigation and recovery towards the rebuilding of quality essential health services. Furthermore, the pulse survey produced globally comparable findings on the extent of disruptions across health systems throughout the COVID-19 pandemic. Essential public health functions were identified across the objectives and aims of the pulse survey, including prominently in public health emergency management, health protection, health promotion in the emergency context, disease prevention and community engagement.

COVID-19 Strategic Preparedness and Response Plan: Global Monitoring and Evaluation Framework. The framework was developed to support monitoring and reporting of global implementation of the COVID-19 Strategic Preparedness and Response Plan *(22)*. The Global Monitoring and Evaluation Framework enabled analysis of the response process to inform adjustments to health policy, health system operations and decision-making, as well as supporting accountability and transparency across countries. The Strategic Preparedness and Response Plan had two main objectives: to reduce and control the incidence of SARS-CoV-2 infections to protect individuals, especially those in vulnerable situations, from exposure and reduce the risk of future variants and to prevent, diagnose and treat COVID-19 to reduce mortality, morbidity and long-term consequences of infection to a minimum. As such, public health emergency management, health protection and disease prevention were identified in the intended purposes of the tool, with health promotion and community engagement identified as integrated elements in achieving these aims.

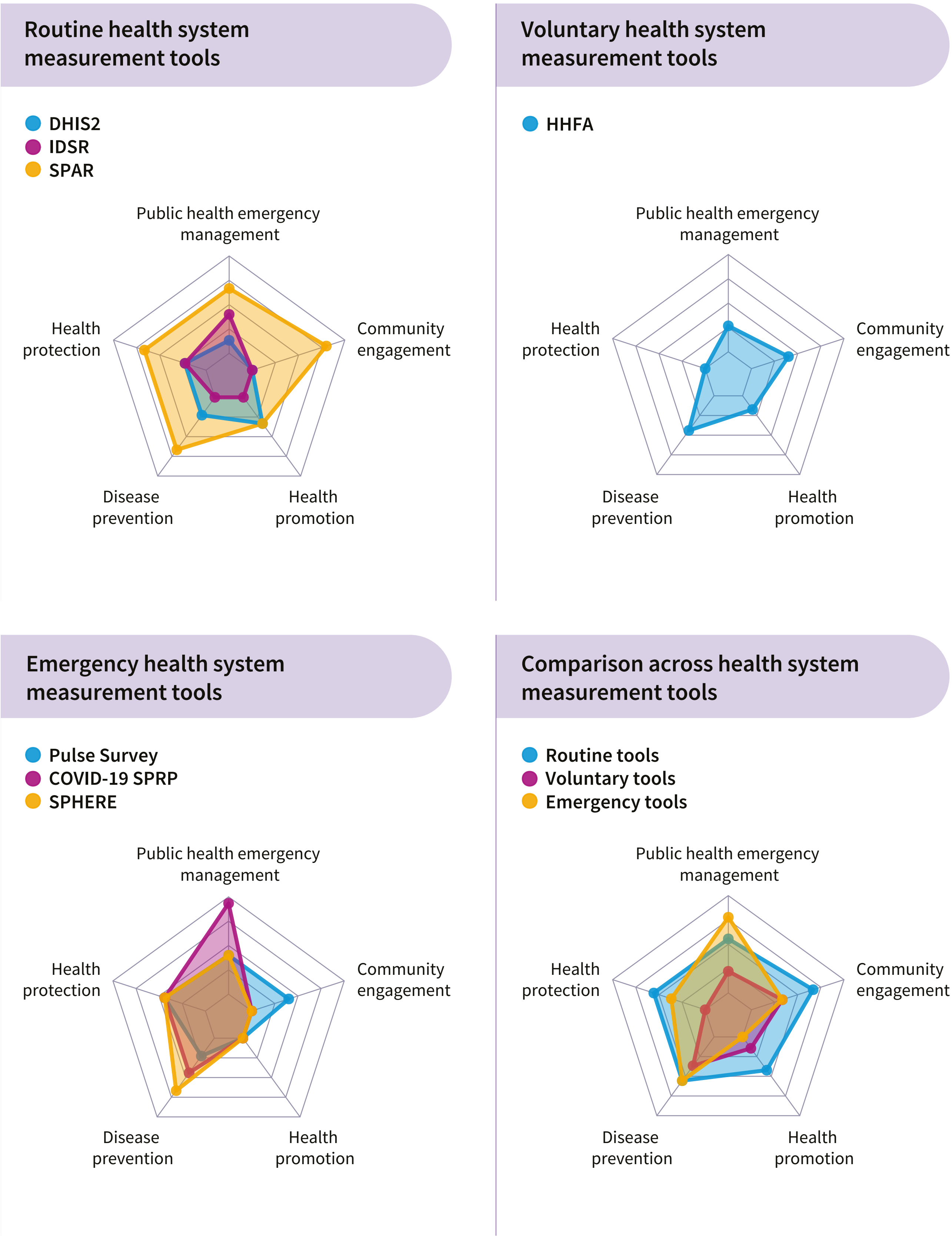
Sphere Handbook. The Sphere Handbook presents a Humanitarian Charter and Minimum Standards in Humanitarian Response *(23)*. Within a human rights-based framework, the Sphere Handbook builds on the legal and ethical foundations of humanitarianism with pragmatic guidance and global good practice. It also compiles evidence to support humanitarian staff wherever they work, including the Sphere Minimum Standards for Healthcare. These standards provide indicators that should be met in every humanitarian context as a fundamental set of requirements for response. All EPHFs are conceptually reflected within the objectives of Sphere standards, with particular emphasis on public health emergency management and health protection, given the emergency nature of the context where they are applied.

Table 4 presents a summary of the tools reviewed.

Table 4. Summary information on tools reviewed

Tool	Purpose	Typical use	Lead or author organization
DHIS2	Open-source platform for health data collection, management and analysis	National health information systems; routine service delivery data; dashboards for decision-making	University of Oslo (Department of Informatics), supported by WHO and global partners
Integrated Disease Surveillance and Response (IDSR)	Strengthens national surveillance and response systems for epidemic-prone diseases	Routine disease surveillance and outbreak reporting at district, national and regional levels	WHO Regional Office for Africa
States Parties self-assessment annual reporting (SPAR)	Monitors country implementation of International Health Regulations (IHR 2005)	Annual country self-assessment of IHR capacities (surveillance, laboratories, risk communication, point of entry)	WHO
Harmonized Health Facility Assessment	Standardized tool to assess service availability, readiness, and quality at health facilities	Comprehensive facility-level assessments feeding into health sector reviews and monitoring of universal health coverage	WHO, with World Bank, United States Agency for International Development, Global Fund to Fight AIDS, Tuberculosis and Malaria, United Nations Children's Fund and other partners
Pulse survey on continuity of essential health services during the COVID-19 pandemic	Rapid monitoring tool to assess service disruptions during crises (especially COVID-19)	Short surveys with ministries of health to track disruptions in essential health services	WHO
COVID-19 Strategic Preparedness and Response Plan: Global Monitoring and Evaluation Framework	Monitors implementation of the COVID-19 Strategic Preparedness and Response Plan	Global and national tracking of the COVID-19 pandemic response actions across health systems, laboratories, supply and risk communication	WHO
Sphere Handbook (Sphere Minimum Standards for Healthcare)	Sets minimum humanitarian standards in disaster and emergency response	Guides humanitarian actors in health, water, sanitation and hygiene, food security, and shelter interventions	Sphere Association (nongovernmental organizations plus International Red Cross and Red Crescent Movement)

Figure 3. Comparison of health system measurement tools in their consideration of essential public health functions



3.2 Comparison of existing health systems and health security assessment tools with integrated measurement of EPHFs

Overviews of coverage for each service-oriented and community engagement public health function measurement and groups of tools are provided in Table 5, with the remaining functions set out in Annex 1. In addition to the summary in Table 5, the coverage is represented diagrammatically in Figure 3, providing a high-level weighted representation of the relative contribution from each tool or set of tools towards measurement of the specified functions. The four indicative measures provide a greater weighting in these representations than partial measures, which represent most contributions to measurement of the respective subfunctions.

The relative contributions of existing tools to measurement of the service-oriented and community engagement public health functions are provided in the diagrams set out in Figure 3. These are categorized across routine, voluntary and emergency health system and health security assessment tools.

Table 5. Cross-mapping of EPHFs and selected health system assessment tools focused on service-oriented functions and community engagement

Note: Partial levels of measurement are shaded yellow, while indicative levels of measurement are shaded green. Grey shading denotes where indicators in the tools do not capture the given public health subfunction.

HHFA = Harmonized Health Facility Assessment; IDSR = Integrated Disease Surveillance and Response: Pulse survey = pulse survey on continuity of essential health services during the COVID-19 pandemic; SPAR = States Parties self-assessment annual reporting; Sphere = Sphere Handbook; SPRP M&E = COVID-19 Strategic Preparedness and Response Plan: Global Monitoring and Evaluation Framework.

		Health system assessment tools						
EPHF	Subfunctions	DHIS2	IDSR	SPAR	HHFA	Pulse survey	SPRP M&E	Sphere
Public Health Emergency Management (EPHF2)	Subfunction 2.1: Monitoring and analysing available public health information to identify and anticipate potential and priority public health risks, including public health emergency scenarios	2.1	2.1	2.1			2.1	2.1
	Subfunction 2.2: Planning and developing capacity for public health emergency preparedness and response as part of routine health system functioning in collaboration with other sectors, including development of a national health emergency response operations plan		2.2	2.2	2.2	2.2	2.2	2.2
	Subfunction 2.3: Carrying out and coordinating effective and timely public health emergency response activities while supporting the continuity of essential functions and services			2.3	2.3	2.3	2.3	2.3
	Subfunction 2.4: Planning and implementing recovery from public health emergencies with an integrated health system strengthening approach					2.4	2.4	
	Subfunction 2.5: Engaging with affected communities and stakeholders in the public and private sectors and health and allied sectors as part of whole-of-government and whole-of-society approaches to public health emergency management		2.5	2.5			2.5	

Table 5 (continued). Cross-mapping of EPHFs and selected health system assessment tools focused on service-orientated functions and community engagement

		Health system assessment tools						
EPHF	Subfunctions	DHIS2	IDSR	SPAR	HHFA	Pulse survey	SPRP M&E	Sphere
Health Protection (EPHF5)	Subfunction 5.1: Developing, implementing, monitoring and evaluating regulatory and enforcement frameworks, including compliance with international legislation, and mechanisms for the protection of specified populations (for example, workers, patients, consumers) and the general public from health hazards			5.1				
	Subfunction 5.2: Conducting risk assessments, risk communication and other risk management actions needed for all manner of health hazards			5.2		5.2	5.2	5.2
	Subfunction 5.3: Monitoring, preventing, mitigating and controlling confirmed and potential health hazards	5.3	5.3	5.3		5.3	5.3	5.3
Disease Prevention (EPHF6)	Subfunction 6.1: Designing, implementing, monitoring and evaluating interventions, programs, services and platforms for primary, secondary and tertiary prevention, including consideration of equity	6.1		6.1	6.1			6.1
	Subfunction 6.2: Integrating consideration of prevention and early detection into service delivery platform design or redesign			6.2	6.2		6.2	6.2
	Subfunction 6.3: Working with partners to support the development, implementation, and monitoring of legislation, policies and program activities aimed at reducing exposure to risk factors and promoting factors that prevent disease			6.3		6.3	6.3	6.3
Health Promotion (EPHF7)	Subfunction 7.1: Designing, implementing and evaluating specific interventions or programs to promote health, including changes in behaviour, lifestyle, practices, and the environmental and social conditions that promote health and reduce health inequities	7.1			7.1			
	Subfunction 7.2: Taking and supporting action, with partners, to address wider determinants of both communicable and noncommunicable diseases through a whole-of-government, whole of-society approach, including increasing individual and community participation in health impacting decisions			7.2				
	Subfunction 7.3: Advocating, developing and monitoring legislation and policies aimed at promoting health and healthy behaviours and reducing inequities	7.3		7.3				
	Subfunction 7.4: Undertaking evidence-based advocacy and health communication to promote healthy behaviours and socioecological environments and build community trust							

Table 5 (continued). Cross-mapping of EPHFs and selected health system assessment tools focused on service-orientated functions and community engagement

		Health system assessment tools						
EPHF	Subfunctions	DHIS2	IDSR	SPAR	HHFA	Pulse survey	SPRP M&E	Sphere
Community Engagement (EPHF8)	Subfunction 8.1: Promoting participatory decision-making and planning for health and the promotion of societal change that enhances, promotes, and protects health and well-being			8.1	8.1			
	Subfunction 8.2: Building community capacity for participating in public health planning, interventions, services, and preparedness and response measures			8.2	8.2	8.2		
	Subfunction 8.3: Monitoring and evaluation of community engagement in public health planning, interventions, services, and preparedness and response measures to promote equity and inclusion			8.3	8.3	8.3		
	Subfunction 8.4: Mobilizing and collaborating with communities and civil society groups in health services, interventions, and programs as part of a whole-of-society approach			8.4		8.4		
	Subfunction 8.5: Engaging communities in health preparedness, readiness, response and recovery			8.5				

3.2.1 Health promotion

Health promotion refers to activities that promote health and well-being and actions that address the wider determinants of health and inequity *(3)*. Routine health system measurement tools provide information towards the measurement of multiple elements of health promotion. DHIS2 provides information relevant to the measurement of subfunction 7.1: Designing, implementing and evaluating specific interventions or programmes to promote health, including changes in behaviour, lifestyle, practices, and the environmental and social conditions that promote health and reduce health inequities, by providing data on mortality, morbidity, service utilization and access alongside service outputs and coverage. SPAR contributes to the measurement of health promotion capacities through subfunction 7.2: Taking and supporting action, with partners, to address wider determinants of both communicable and noncommunicable diseases through a whole-of-government, whole of-society approach, including increasing individual and community participation in health-impacting decisions, by assessing C10: Risk communication and community engagement, and C13: Food safety.

DHIS2 and SPAR both provide information relevant to subfunction 7.3: Advocating, developing, and monitoring legislation and policies aimed at promoting health and healthy behaviours and reducing inequities. Routine health system measurement tools were not found to provide information on the measurement of subfunction 7.4: Undertaking evidence-based advocacy and health communication to promote healthy behaviours and socioecological environments and build community trust.

The voluntary health system measurement tool Harmonized Health Facility Assessment contributes to the measurement of health promotion capacities reflecting subfunction 7.1: Designing, implementing and evaluating specific interventions or programmes to promote health, including changes in behaviour, lifestyle, practices, and the environmental and social conditions that promote health and reduce health inequities, through dimension 5: Clinical quality of care. Existing emergency health system measurement tools were not found to provide information relevant to measuring health promotion subfunctions.

3.2.2 Health protection

Health protection consists of activities that protect populations against health threats, for example, environmental and occupational hazards and communicable and noncommunicable diseases, including mental health conditions, food insecurity, and chemical and radiation hazards *(3)*. In the measurement of EPHFs, this is primarily considered in routine rather than emergency management settings. Routine health system measurement tools DHIS2, IDSR and SPAR all contribute to the measurement of health protection capacities through providing data on monitoring, preventing, mitigating, and controlling confirmed and potential health hazards (subfunction 5.3).

SPAR further contributes to the measurement of health protection capacities, as C1: Policy, legal and normative instruments to implement IHR provide information useful to measure health protection capacity – specifically subfunction 5.1 on developing, implementing, monitoring and evaluating regulatory and enforcement frameworks, including compliance with international legislation, and mechanisms for the protection of specified populations (for example, workers, patients, consumers) and the general public from health hazards. SPAR provides a wealth of information on the capacity for conducting risk assessments, risk communication and other risk management actions needed for all manner of health hazards (subfunction 5.2) through C5: Surveillance, C7: Health emergency management, C10: Risk communication and community engagement, C11: Points of entry and border health, and C13: Food safety.

As IHR *(2005)* is the foremost international legislative instrument for health hazards, measurement through SPAR in this area can be considered to cover most of this aspect of health protection public health capacity, albeit focused on infectious, chemical and radiation threats of public health importance. Some domestic health hazards that are not of wider concern may not be covered in the SPAR tool, which is designed to assess compliance with IHR.

The voluntary health system measurement tool Harmonized Health Facility Assessment provides information through “general service readiness indicators” on standard precautions for infection prevention, which support the measurement of subfunction 5.1: Developing, implementing, monitoring and evaluating regulatory and enforcement frameworks, including compliance with international legislation, and mechanisms for the protection of specified populations (for example, workers, patients, consumers) and the general public from health hazards. The Harmonized Health Facility Assessment also provides information on emergency preparedness and resilience, which covers all areas of health protection capacities and is integrated with public health emergency management. The Harmonized Health Facility Assessment is focused on the facility level, so it does not provide a full picture of health protection capacities beyond the facilities, which should also be considered to assess full public health capacity in this area.

Emergency health system measurement tools provide information relevant to measuring subfunctions 5.2 and 5.3. The pulse survey on continuity of essential health services during the COVID-19 pandemic provides this information, particularly through the area of future acute respiratory pandemic preparedness. The Global Monitoring and Evaluation Framework of the COVID-19 Strategic Preparedness and Response Plan supports the measurement of health protection capacities for monitoring, preventing, mitigating and controlling confirmed and potential health hazards as part of measuring public health intelligence, risk forecasting, response monitoring and health system responses. The Global Monitoring and Evaluation Framework also provides information on the measurement of conducting risk assessments, risk communication and other risk management actions needed for all manner of health hazards, particularly in the areas of risk communication and infodemic management.

Emergency health system measurement tools provide less information on subfunction 5.1 (regulatory and enforcement frameworks), given that such frameworks may not be functional in emergency and humanitarian contexts.

3.2.3 Disease prevention and early detection

Disease prevention refers to activities related to the prevention and early detection of communicable and noncommunicable diseases, including mental health conditions and injuries *(3)*. In the measurement of EPHFs, this is primarily considered in routine rather than emergency management settings. Routine health system measurement tools, including DHIS2 and SPAR, provide information towards the measurement of the public health capacities in disease prevention, particularly the subfunction 6.1 element on monitoring and evaluating interventions, programmes and services for primary, secondary and tertiary prevention. SPAR also includes criteria of use in measuring subfunction 6.2 and subfunction 6.3, particularly in areas of infection prevention and control (C9) and food safety (C13). Importantly, SPAR also considers the legislative and policy environment for reducing exposure to disease risk factors (C1, C12). IDSR is focused on disease surveillance and response, so it is not directly able to measure disease prevention capacities as delineated in the EPHFs.

The voluntary health system measurement tool Harmonized Health Facility Assessment provides information on the availability of preventive public health services covering subfunction 6.1 and subfunction 6.2, particularly those provided through health facilities. It also provides information on how prevention and early detection have been integrated into health service delivery across a range of major disease areas for prevention, including antenatal care, HIV and other communicable diseases.

Emergency health system measurement tools, including the pulse survey on continuity of essential health services during the COVID-19 pandemic, provide information on the legislative environment for disease prevention (subfunction 6.3) as it relates specifically to COVID-19, so it may be of limited benefit to wider public health capacity measurement. The Global Monitoring and Evaluation Framework of the COVID-19 Strategic Preparedness and Response Plan provides valuable information towards the measurement of subfunction 6.2 and subfunction 6.3 on disease prevention capacities in relation to infection prevention and control. Sphere provides information that can support assessment across disease prevention capacities (subfunctions 6.1, 6.2, 6.3) in a number of essential services, including HIV, vaccine-preventable diseases, and reproductive, maternal and newborn health care.

3.2.4 Public health emergency management

Public health emergency management refers to managing public health emergencies of cross-border and global security concern *(3)*. The routine health security measurement tool SPAR is well suited to the measurement of the integrated delivery of public health emergency management, in particular:

- C5 (Surveillance), which correlates well with subfunctions of EPHF on public health emergency management focused on monitoring and analysing available public health information to identify and anticipate potential and priority public health risks, including public health emergency scenarios.
- C7 (Health emergency management), which correlates well with subfunctions of EPHF on public health emergency management focused on planning and developing capacity for public health emergency preparedness and response as part of routine health system functioning in collaboration with other sectors, including development of a national health emergency response operations plan.
- C8 (Health services provision), which correlates well with subfunctions of EPHF on public health emergency management focused on carrying out and coordinating effective and timely public health emergency response activities while supporting the continuity of essential functions and services.
- C10 (Risk communication and community engagement), which correlates well with subfunctions on communicating public health data, information and evidence with key stakeholders, including communities.

As part of the health information system in many countries, DHIS2 provides additional information that can be used to identify and anticipate potential and priority public health risks, including emergencies. IDSR also provides a wide spectrum of information that supports integrated measurement of public health emergency capacities across monitoring, response, coordination and community engagement.

The voluntary health system measurement tool Harmonized Health Facility Assessment provides information relating to emergency preparedness and resilience of health facilities (dimension 2.8) and governance and management systems (dimension 4.1), which are of relevance to the measurement of delivery of public health emergency management.

Emergency health system measurement tools, including the pulse survey on continuity of essential health services during the COVID-19 pandemic, provide information on public health emergency management, albeit limited to the specific context for which the pulse survey was designed. They cover assessment of the public health emergency response while supporting the continuity of essential functions and services and recovery from public health emergencies across a core range of essential health services. The Global Monitoring and Evaluation Framework of the COVID-19 Strategic Preparedness and Response Plan covers elements of public health emergency management, including health system recovery planning and monitoring of disruption of essential services. The Sphere Handbook provides information on public health emergency management in humanitarian contexts, specifically on the provision of health services in emergencies (standard 1.1), surveillance and outbreak detection (standard 2.1.2), outbreak preparedness and response (standard 2.1.4) and injury and trauma care (standard 2.4) for mass casualty situations.

3.2.5 Community engagement and social participation

Community engagement and social participation involve strengthening community engagement, participation and social mobilization for health and well-being, with respect for the culture of the people *(3)*. The routine health system and health security measurement tool SPAR is well suited to the measurement of community engagement in relation to health security threats, in particular C10 (Risk communication and community engagement), which correlates well with subfunctions of EPHF on communicating public health data, information and evidence with key stakeholders, including communities, and engaging communities in health preparedness, readiness, response and recovery. IDSR can provide information of use in the assessment of this public health function under the component on monitoring, evaluation, supervising, and providing feedback to improve surveillance and response (section 8). DHIS2 does not generally provide routine information to assess community engagement as a public health function.

The voluntary health system measurement tool Harmonized Health Facility Assessment provides some information to measure community engagement as a public health function through dimension 4.1: Governance and management systems, which support participatory decision-making and planning for health, assessing community capacity for participating in public health planning, and monitoring and evaluation of community engagement.

Emergency health system measurement tools, including the pulse survey on continuity of essential health services during the COVID-19 pandemic, provide information on mitigation and recovery measures (within the continuity of essential health services module), which could have supported the measurement of the delivery of community engagement in public health functions. The Global Monitoring and Evaluation Framework of the COVID-19 Strategic Preparedness and Response Plan and Sphere Handbook do not provide significant detail on measurement of community engagement capacities.

3.3 Integration in measurement across EPHFs of existing health system and health security measurement tools

In addition to the specific comparison of each set of indicators with all EPHFs, an assessment was made of where assessment tools represented integrated measures across more than one EPHF.

The routine health system and health security measurement tool SPAR considers risk communication and community engagement (C10) in the context of public health emergency management (subfunction 2.5) and health protection (subfunction 5.2). At its highest level, this is defined as the presence of mechanisms for systematic community engagement in public health emergencies, including guidelines and standard operating procedures, which have been developed, disseminated, and implemented, and supported at national, intermediate, and local levels. To achieve this level, a country must also conduct qualitative and quantitative sociobehavioural research and have mechanisms and activities for community engagement exercised, reviewed, evaluated and updated on a regular basis. As SPAR is a self-assessment reporting tool, the level of functioning is determined by national-level authorities with additional processes, including the Joint External Evaluation tool providing external support to these.

In the Harmonized Health Facility Assessment, emergency preparedness and resilience (dimension 2.8) indicators also consider both public health emergency management (subfunctions 2.2, 2.3) and routine health protection scenarios (subfunctions 5.1,.5.2, 5.3). SPAR also provides information on health promotion (subfunctions 7.2, 7.3) in the context of public health emergency management and routine health protection scenarios through policy, legal and normative instruments to implement IHR (C1), risk communication and community engagement (C10) and food safety (C13) capacity measurement. At present, this information is focused on communication and public engagement.

The pulse survey on continuity of essential health services during the COVID-19 pandemic provides information on community engagement and social participation (subfunctions 8.2, 8.3 and 8.4) within the context of public health emergency management (subfunctions 2.2 and 2.4) as part of the mitigation strategies and recovery measures set of indicators. The pulse survey also considers disease prevention (subfunction 6.3) within the context of public health emergency management as part of the policies, planning and investments set of indicators.

In emergency health system measurement, the Sphere Handbook was noted to incorporate disease prevention and behaviour change measures (Hygiene promotion, standard 1.1) within the context of ongoing public health emergency management.

4. Discussion

This review covered the seven most widely used monitoring tools that are available from country to global level. Routine, voluntary and emergency measurement tools were found to provide valuable information on a wide range of public health functions and services consistent with their defined scope. Areas remaining as gaps in measurement were identified and should be addressed with additional or revised indicators and guidance to ensure a more comprehensive approach to measuring EPHFs across routine and emergency contexts. A need was also identified for further integrated measurement of EPHFs to promote the alignment of and accountability for public health across functional areas. These are discussed below with a focus on health promotion, disease prevention, health protection and public health emergency management, and community engagement.

4.1 Health promotion

Public health emergencies, including the COVID-19 pandemic, have highlighted the need for health promotion as a tool to address health crises and to improve the health of populations, particularly those experiencing deprivation, marginalization or displacement *(24)*. The Ottawa Charter for Health Promotion *(1986)* highlighted the centrality of health promotion to the health and well-being of populations and a shared responsibility for health and allied sectors *(7)*. In this review, health promotion measurement was observed in routine and voluntary health system measurement tools, focused on disease-specific measures, interventions and programmes to promote health with a focus on the health sector. More gaps were identified in measurement of health promotion capacities in emergency-related measurement tools, although the pulse survey on continuity of essential health services during the COVID-19 pandemic may have indirectly assessed some health promotion through measurement of tracer service areas, including HIV, noncommunicable diseases and substance use disorders. Evidence-based advocacy and health communication within health promotion was a notable gap (subfunction 7.4) that was not addressed by existing tools and represents an opportunity for further specific measurement approaches. This is consistent with documented lessons from emergencies such as the COVID-19 pandemic; for example, the 2023 report of the Chief Public Health Officer of Canada highlighted the importance of health promotion in strengthening emergency management by working with communities to advance health equity and foster social cohesion *(25)*.

4.2 Health protection

Information on identifying, monitoring, preventing, mitigating and controlling confirmed and potential health hazards is widely available from existing health system measurement tools and can be used across contexts to determine the level of public health capacity in this area. It is important to note that not all health hazards may be included in each tool, so a combination of available indicators may be required to ensure that an all-hazards approach is provided, and care must be taken not to assume that a high level of coverage for some hazards (such as infectious disease or radiological hazards) is reflective of full capacity in this area. No single indicator set was found to provide a fully indicative measure of health protection capacities, while IHR *(2005)* SPAR and Joint External Evaluation are designed to assess the capacity for responding to various hazards from infectious outbreaks to chemical and radiation release. It is also important to note that although compliance with routine tools that provide information on health protection capacities, such as SPAR, is almost universal globally at national level, areas that are experiencing fragility or protracted emergencies and requiring humanitarian assistance may need supplementary measurement of health protection functions. Additional gaps were found in relation to consideration of the full range of specified at-risk populations for protection and ensuring that an all-hazards approach is adopted from local to national level, including regulatory capacities.

4.3 Disease prevention and early detection

Wide and varied information was found to be available from existing health system measurement tools to support the measurement of delivery of disease prevention and early detection, from disease- and service-specific indicators (particularly at facility level) to the stakeholder and community engagement needed to support legislation, policies and programme activities that prevent disease. Information was limited on primary prevention capacities and disease prevention activities beyond the health sector. No single indicator set was found to provide a fully indicative measure of disease prevention and early detection capacities, with the monitoring of legislation and policies related to disease prevention noted as a significant area for further measurement. In addition, disease prevention measures assessed do not routinely embed equity as a central consideration, representing an area for additional specific measurement.

4.4 Public health emergency management

Existing measurement tools provide more information on the measurement of public health emergency management capacities than any other EPHF. This is probably due to an increased global focus on measurement in this area through the requirement for countries to report annually through SPAR or voluntarily every five years through Joint External Evaluation. Routine measurement tools provide information that can be used directly to assess the measurement of emergency management, albeit focused heavily on specific types of public health emergencies – predominantly infectious disease outbreaks and chemical, biological or radiation events. Furthermore, tools, including IDSR, cover much of public health emergency management but do not aim to provide information on recovery, and provide little on continuity of essential health services as part of emergency management. More recently developed tools (such as the second edition of SPAR) and those focused on emergencies provide valuable information on recovery and engagement of communities in emergency response. Countries may look to collect specific pieces of information to supplement the assessment of public health emergency management capacities, but these are likely to be quite focused on specific gaps, for example, the capacity to manage a rapid increase in noncommunicable disease necessitating an emergency response, stakeholder engagement and community voice, or recovery planning and implementation.

4.5 Community engagement and social participation

Community engagement and social participation information provided by existing health system measurement tools is strongly focused on the contexts of outbreak management and broader health security, where there have been major efforts to recognize and monitor the inclusion and agency of communities in supporting the management of diseases affecting their health. This was notable during the COVID-19 pandemic, where concerted efforts were made to enshrine community engagement with marginalized and underrepresented communities in Canada and similar contexts *(26)*. This is reflected in tools such as SPAR, which can be used directly to measure capacities, including engagement of communities in health preparedness, readiness, response and recovery. The voluntary tool Harmonized Health Facility Assessment also includes indicators that reflect community engagement in health services, and the COVID-19 pandemic further provided opportunities to embed measurement of community engagement in the public health emergency response *(27)*. There are gaps in the measurement of community engagement and social participation in routine health systems and public health planning and delivery, for which additional measurement would be valuable. Community engagement is not yet fully considered in general emergency and humanitarian contexts, which is needed to provide adequate coverage of all EPHFs in all contexts.

4.6 Comprehensive approach to measurement of EPHFs

Existing health system measurement tools provide valuable information on the measurement of EPHFs, but the use of already embedded tools alone is not sufficient to enable a fully comprehensive assessment. However, currently, the focus of health monitoring is siloed: some monitoring tools are dedicated to measuring health security capacity from a national to a cross-border perspective, while others relate to the availability of health services from a service coverage perspective. There is currently no tool that allows an integrated approach to health sector performance for both universal health coverage and health security with due consideration of intersectoral coordination and responsibility for public health. To enable holistic measurement across the EPHFs, a combination of existing data sources, indicators and newly collected and categorized information is needed, informed by systematic cross-mapping between available data and the EPHFs, considering the more granular subfunctions as applicable.

Existing tools and platforms can be adapted and expanded to cover gaps in public health capacity measurement identified, for example, DHIS2 can be developed to cover a broader range of public health functions, including health promotion and community engagement activities, rather than being focused only on health outcomes and health system metrics. As routine use of DHIS2 continues to expand, this represents an opportunity for the systematic embedding of measurement of further aspects of public health delivery. Health facility-focused assessments already provide information on the public health role of primary and secondary care, but a significant opportunity exists to extend such assessments to further cover disease prevention and health promotion activities. As emergency health system measurement tools develop, for example in future surveys based on the pulse survey approach, broader consideration should be given of health promotion as a key element of public health emergency response.

Some gaps remain across all existing measurement tools, for example health promotion subfunction 7.4, which includes evidence-based advocacy and health communication. Creation of stand-alone tools is costly and difficult to administer, but well targeted additional measurement is needed to assess countries' capacities to deliver in this area, which should be accompanied by existing data to provide a complete picture.

4.7 Integrated measurement of EPHFs

An integrated approach to measurement supports the development of people-centred public health services and the enabling functions. Community engagement represents a valuable focus for enhancing integrated measurement, given the importance of context-appropriate community-oriented approaches to sustainability and resilience of public health systems.

Aspects of existing tools were found to provide integrated measurement of EPHF subfunctions, but none of the reviewed tools was found to fully capture the integrated and comprehensive measurement of the 12 EPHFs, demonstrating that further integrated measurement is required. Significant areas of good practice were found that demonstrate the value of existing tools to support integrated measurement. For example, in routine contexts, the SPAR tool provides information on the integrated coverage of public health emergency management, community engagement in emergencies and health protection functions. The Global Monitoring and Evaluation Framework of the COVID-19 Strategic Preparedness and Response Plan includes information to measure major areas of public health emergency management and health protection functions. The Sphere standards provide valuable information integrating emergency management, health protection and disease prevention in humanitarian contexts. It is acknowledged that not every indicator can represent integration across the EPHFs, and a specific focus is appropriate where tools have a defined priority or purpose; for example, the pulse survey was specifically looking at existing services rather than planning for new services.

There is a need for indicators that not only cover more than one EPHF in tandem but that truly capture integration and synergies among EPHFs. Such indicators can promote integration by bringing together areas of public health work, for example, where stakeholder and community engagement is more central to emergency management. They can also enable and facilitate the involvement of communities in routine public health promotion delivery.

It is recommended that indicators be developed that provide measurement of integration across EPHFs. Examples of such indicators are included in the health system resilience indicators package *(28)*, which provides integrated measures across public health functions, for example, the presence of a health system focal point for IHR reporting. Development of such indicators can address areas where there is a lack of integration in existing measurement tools.

4.8 Strengths and limitations of review

This technical assessment provides a systematized, objective and detailed review of the existing health systems and health security measurement tools and their adequacy in providing a measurement of the EPHFs, including integrated measurement. As such, it supports the comprehensive development of public health functions in all contexts and demonstrates how countries already capture much of the information they need to determine their public health capacities.

This report focused on a selected number of health systems and health security measurement tools, and countries may have additional tools and adaptations in addition to the publicly accessible tools included here. In assessing the adequacy of existing tools against the EPHFs, a determination was made of partial and indicative coverage of each function, to support a consistent approach across diverse functions and tools. This delineation does however represent a simplification of the variety of levels of coverage, and it should not be concluded that all “partial” measures are of the same level of coverage. This report focused specifically on service-oriented EPHFs, which require integrated delivery with system enabling-oriented and cross-cutting EPHFs. Further review should focus on these system enablers and be considered in tandem to effectively strengthen all public health functions together.

Part B

5. Way forward and entry points for action

This section provides guidance for improving the integration of measurement of EPHFs in existing health systems and health security tools and processes in countries. The recommendations are organized as follows:

- actions and entry points at the measurement level for reorienting existing health system measurement tools towards the integrated measurement of EPHFs and services;
- enabling and policy-focused actions at governance level for promoting integration and reducing fragmentation across health and associated sectors;
- a concise list of example indicators that countries and other stakeholders can select from, adapt and utilize for enhancing the measurement of EPHFs using an integrated approach;
- operational steps with example scenarios, country examples and a checklist to guide country-level implementation.

5.1 Actions at measurement level

The following actions are recommended at the measurement level.

1. Expand metric development beyond the health sector

Ongoing and future metric development – whether routine, ad hoc or emergency focused – should include indicators capturing health promotion and protection functions located outside the health sector (for example, social care by local administrations, environmental health, water and food safety).

2. Strengthen measurement of health promotion in emergencies

Specific measures should be designed to assess comprehensive health promotion capacities within emergency management contexts, leveraging and integrating with routine health information systems.

3. Develop targeted indicators for evidence-based advocacy and health communication (subfunction 7.4)

Indicators should be designed or integrated into existing health system monitoring tools to measure activities related to evidence-based advocacy, health communication, and community trust building for healthy behaviours and socioecological environments.

4. Adopt an all-hazards approach to capacity measurement

Information from existing measurement tools should be analysed in combination and supplemented to ensure comprehensive coverage of all hazards – natural, technological and societal – within health protection capacity assessments.

5. Integrate equity and inclusion across all capacity measurements

Indicator development for health protection and disease prevention should explicitly incorporate equity and inclusion considerations, ensuring that all at-risk, marginalized and minoritized populations – such as Indigenous Peoples, migrants and communities experiencing stigma – are adequately represented. Measurement frameworks should assess not only the existence of such activities but also the extent to which they reach and benefit these populations, ensuring comprehensive and equitable coverage across all groups.

6. Integrate cross-sectoral disease prevention indicators

Existing health system measurement tools should incorporate indicators that capture disease prevention capacities beyond the health sector, including community-based and outreach initiatives focused on primary prevention.

7. Track recovery and resilience building in health systems

Additional measurement efforts should monitor the planning and implementation of health system recovery processes using an integrated health system strengthening lens, providing a fuller understanding of emergency management capacity.

8. Measure community engagement and social participation

Indicators should be developed or expanded to capture community engagement and social participation in routine public health and health system planning, building on existing health system resilience indicator packages.

9. Adapt humanitarian health system tools for participation and equity

Health system assessment tools applied in humanitarian or fragile contexts should be refined to include metrics that assess capacities for community engagement and social participation, tailored to local population health needs.

5.2 Actions at governance level

The following policy actions are recommended to support and enable improvements in measuring EPHFs.

1. Review and strengthen national measurement frameworks through an EPHF lens

Authorities responsible for national health information system, ad hoc, and emergency-related data collection should commission a review of existing measurement frameworks and indicators in collaboration with relevant stakeholders. This review should apply the EPHF lens to identify areas for improvement based on population health needs and public health priorities. Where appropriate, a multidisciplinary and multisectoral working group should be established with a clear mandate to ensure comprehensive and integrated measurement of public health capacities and to facilitate data integration across sectors.

2. Integrate EPHFs into national health and health security policies and plans

Health sector and health security policies and plans – such as national health sector strategies and the national action plan for health security – should explicitly incorporate the full range of EPHFs when defining priorities, actions, and monitoring and evaluation frameworks. This approach helps prevent fragmentation and promotes coherence between universal health coverage and health security agendas.

3. Develop workforce competencies in measurement and monitoring across EPHFs

The competencies of the public health workforce, as outlined by WHO, should be strengthened to enable countries to review, develop, select and apply measurement frameworks and indicators that generate context-appropriate data across all EPHFs.

Capacity-building efforts may include pre-service and in-service training, mentorship, and peer-to-peer learning initiatives within and between countries.

4. Align donor and partner investments with integrated EPHF measurement and delivery

Donors and development partners providing financial or technical support should promote and incentivize integrated approaches to public health function delivery and measurement based on population health needs. This alignment can be achieved through funding requirements, strategic guidance and technical assistance frameworks that emphasize EPHF coherence.

5. Update and harmonize measurement tools across routine and emergency contexts

Countries should update and adapt existing measurement tools to ensure comprehensive coverage of public health functions and services at all levels of the health system – national, subnational and community. Opportunities such as health sector reforms, post-crisis recovery, and resilience-building initiatives can be leveraged to strengthen public health information systems.

6. Prioritize public health capacity strengthening in all investments and plans

National and subnational authorities should embed public health capacity strengthening into development and humanitarian policies, plans and investments. Existing health system measurement tools should be used and supported as key resources for tracking progress and ensuring accountability.

7. Invest in community engagement and system-enabling public health functions

Governments and partners should invest in community engagement and other enabling public health functions that enhance the delivery of health protection, promotion, disease prevention and emergency management across all contexts. This includes establishing mechanisms for sustained social participation and community partnership in public health planning and implementation.

5.3 List of potential indicators

To support countries in applying the above sets of recommendations, the following indicators (Table 6) are proposed for selection and adaptation based on contextual considerations, including identified gaps in embedding EPHF considerations in the existing measurement approaches and tools. These indicators are not intended to replace the existing indicators being measured through various tools; rather, they are to complement and address gaps in the available routine, ad hoc and health security-related measurement efforts at national and subnational levels. They are informed by published technical resources, including WHO's health system resilience indicators *(28)* and outcomes of consultations with relevant experts. The proposed indicators are grouped under two categories – those that support EPHF measurement directly and those that measure the enabling institutional capacities at policy and planning levels. This categorization is aligned with and supports measurement of the two composite indicators on EPHFs within the WHO package of health system resilience indicators, that is, the current state of EPHF delivery ascertained and institutional capacity for coordination of EPHFs, as well as outcome indicator in the WHO Fourteenth General Programme of Work 2025–2028 on strengthening institutional capacity for EPHFs, as agreed at the Seventy-seventh World Health Assembly *(6)*.

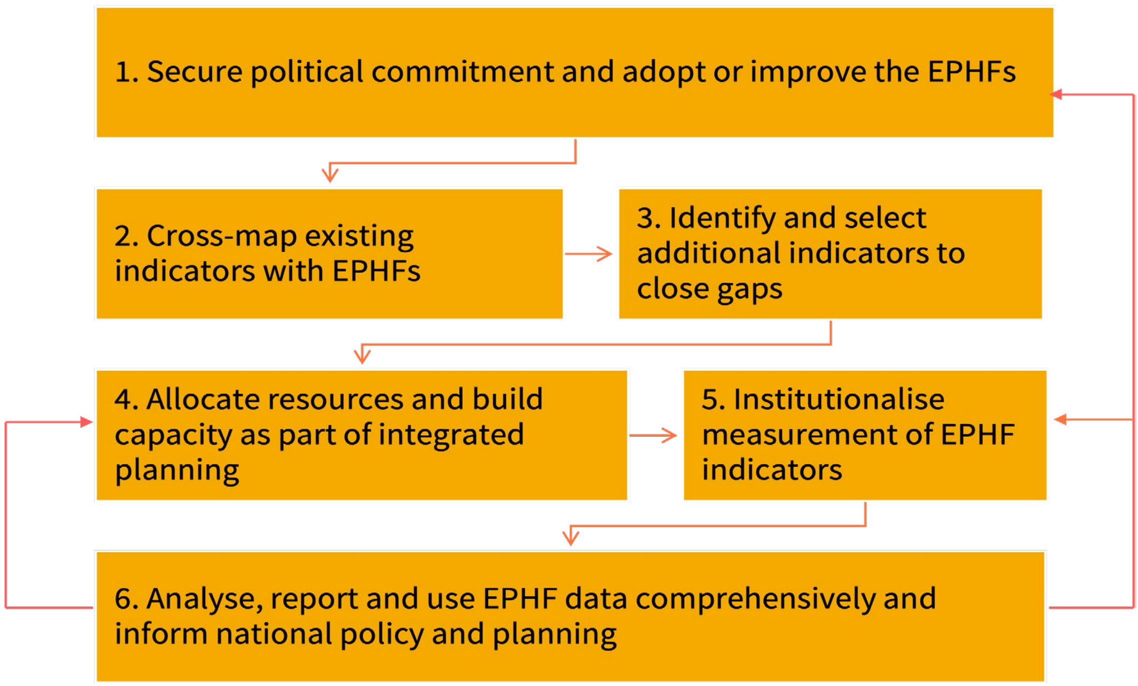
5.4 Stepwise process for integrating EPHF measurement in health systems

The steps described below (and as outlined in Figure 4) are intended to guide national authorities, particularly ministries of health, departments responsible for health information systems, and national public health institutes, in systematically integrating EPHFs into mainstream monitoring and evaluation. These steps assume that political commitment and policy backing are in place to enable integration at all levels. Examples of country-level entry points and scenarios are provided for each step. The suggested entry points indicate examples of who should be engaged, what systems or policies to leverage, and where the work can be anchored.

Table 6. Examples of complementary indicators

Examples of indicators or assessment questions to ensure integrated and comprehensive measurement of EPHF delivery	Examples of indicators of institutional capacity at policy and planning levels for enabling EPHF delivery and measurement
1. Current state of delivery of EPHFs ascertained, considering those within and beyond the health sector 2. Recent (in the past three years) review of consideration of EPHFs in current measurement frameworks and indicators 3. Existing information systems and tools used in emergency management (including humanitarian response) include health promotion, prevention, and protection and community engagement activities 4. National routine health information system covers all EPHFs to be delivered at national, subnational and service delivery levels 5. Health information analysis and reporting for informing policies and plans and resource allocation prioritize all EPHFs, including service-oriented EPHFs and community engagement 6. Health programmes and policies are developed and implemented with documented community-level input	1. Availability of a legal and policy framework for delivery of EPHFs in routine and emergency contexts 2. Existence of a national entity or structure that is responsible for multisectoral coordination of the EPHFs 3. Public health emergency management plans and structures (for example, incident management system, emergency operations centre) include pillars on continuity of routine health services (clearly identifying preventive, promotive and protective services as part of emergency management) 4. Public health emergency preparedness and response plans identify and prioritize the needs of people in vulnerable situations 5. Availability of strategy for developing competencies of public health workforce in EPHFs as appropriate to their roles 6. Package of essential health services to be delivered in all contexts includes all service-oriented EPHFs to be delivered at all levels of care, including primary care

Figure 4. Key steps for integrating EPHFs in existing monitoring processes



5.4.1 Examples of key actions and entry points

Step 1. Secure political commitment and adopt or improve the EPHFs

Examples of key actions

- Advocate high-level political endorsement of the EPHFs as a foundation for integrated strengthening of public health capacities in support of universal health coverage and health security objectives.
- Facilitate national consultations to contextualize and adapt the EPHF framework, ensuring coherence with existing health sector strategies, governance structures and national health priorities.
- Develop and institutionalize a formal mechanism – for example, through a ministerial directive, policy framework or strategic roadmap – with clearly defined leadership and coordination roles to guide integration of EPHFs into relevant multisectoral planning, budgeting, services and monitoring at national and subnational levels.
- Institutionalize periodic strategic reviews (for example, every three to five years) to assess the relevance, coverage and performance of implementation of EPHFs, drawing on evidence from system assessments, after-action reviews and Joint External Evaluation for IHR (2005).

Expected outputs

- Formal political endorsement with integration of EPHFs within national health policies, strategies, investment frameworks, multisectoral collaboration and coordination mechanisms for public health.
- Defined institutional responsibilities for leadership, coordination and oversight of implementation of EPHFs (for example, within and across the ministry of health, national public health institute or intersectoral platforms).
- Established mechanism for regular strategic review ensuring that EPHF implementation remains responsive to evolving health challenges, reforms and lessons learned.
- Enhanced national ownership, visibility and accountability for EPHFs as a core requirement for building resilient, people-centred health systems, contributing to attainment of universal health coverage, health security and the Sustainable Development Goals.

Table 7 presents examples of country-level entry points for step 1, while Box 2 presents an example scenario for implementation of step 1 at country level.

Box 2. Example scenario for implementation of step 1

Country X demonstrates a strong political drive towards systemwide strengthening of public health functions as part of reforms to enhance universal health coverage, health security and attainment of Sustainable Development Goal targets.

The ministry of health, supported by the national public health institute and key partners, initiates a process to adopt and institutionalize the EPHF framework formally.

A high-level multisectoral task force is convened to adapt the global EPHF framework to the country’s context, linking it to existing health sector priorities, governance mechanisms and performance indicators.

The adapted EPHFs are then incorporated into the national health strategy and medium-term expenditure framework, ensuring that public health capacities are embedded in planning and financing cycles.

To maintain political visibility and accountability, the task force institutionalizes periodic strategic reviews conducted every three years using data from health system assessments and public health performance reports. This process enables continuous refinement of EPHF priorities and ensures that political commitment translates into sustained institutional practice and measurable public health gains.

Table 7. Country-level entry point examples for step 1: Secure political commitment and EPHF adoption with periodic strategic review

Key country-level references to anchor and institutionalize EPHF indicators	Relevance
Ministry of health	Serves as lead advocate for EPHF integration; responsible for developing and endorsing policy or strategic frameworks anchoring EPHFs within national health system reforms
National public health institute (or equivalent)	Acts as a technical lead on defining and operationalizing EPHFs in line with its mandate; ensures alignment between EPHFs and national disease control, emergency preparedness and surveillance frameworks
Parliamentary health committee or cabinet-level health council	Provides formal political endorsement and resource allocation for implementation of EPHFs through national policy, legislation or budgetary instruments
Ministry of finance	Aligns EPHF priorities with national development and financing frameworks, ensuring sustainability and domestic resource mobilization
Development partners (for example, WHO, World Bank, bilateral agencies)	Support policy dialogue, facilitates technical adaptation, and mobilize financing for initial implementation and institutionalization processes
Civil society and academic institutions	Contribute to policy advocacy, evidence generation and accountability mechanisms for sustained integration and review of EPHFs

Table 8. Country-level entry point examples for step 2: Cross-map existing indicators with EPHFs

Key country-level references to anchor and institutionalize EPHF indicators	Relevance
Ministry of health (for example, planning, monitoring and evaluation, health information division)	Custodian of national health information systems and accountable for aligning EPHF monitoring with health sector plans; convener for mapping exercise
National public health institute (or equivalent) roles	Technical lead on public health functions, such as surveillance and public health data, emergency preparedness and response, ensuring that mapping reflects the EPHFs led or coordinated by the national public health institute
National health information system platforms (DHIS2, IDSR, SPAR, vertical programme dashboards)	Primary sources of existing indicators and data flows to be mapped against EPHFs
National statistical office	Links health data with broader demographic and socioeconomic indicators (population-based surveys, censuses, national health accounts)
Technical partners (for example, WHO) and academia (for example, public health schools)	Provide expertise to design mapping matrices, ensure methodological rigour, and train staff
Technical working group for monitoring and evaluation, health information system steering committees	Existing coordination mechanisms that bring ministry of health, partners and donors together for indicator alignment

Box 3. Example scenario for implementation of step 2 (continued from step 1)

Country X commits to strengthening its health system capacity for public health through an integrated public health approach, as evidenced by its policies and plans. As part of the implementation efforts, the national health authorities (for example, the ministry of health or national public health institute) are willing improve measurement and monitoring of EPHFs through existing health information mechanisms, in collaboration with other sectors responsible for public health, A working group is established comprising public health and data experts across departments of the ministry of health, the national public health institute and partner organizations.

The responsible working group decides to focus on nationally monitored indicators. They collate the tools and indicators currently monitored at the national level and develop the template below for mapping EPHFs (referencing the subfunctions or services according to WHO's guidance) across indicators from the currently used monitoring tools and mechanisms. (Note: This template can be adapted as needed, for example, according to different levels (national, subnational), frequency or completeness of data collection.)

Sample template: Mapping of national-level indicators across the EPHF subfunctions

EPHFs (WHO's unified list, context-specific adaptation of the list)	Public health subfunctions (or public health service areas)	Corresponding indicators from monitoring tool 1 (e.g. IDSR)	Corresponding indicators from monitoring tool 2 (e.g. DHIS2)	Corresponding indicators from monitoring tool 3 (e.g. SPAR)	Corresponding indicators from monitoring tool 4 (e.g. externally funded primary health care initiative)

Country X finds that data on EPHFs exists across multiple platforms (DHIS2, emergency reporting tools, vertical disease programme dashboards). However, no mechanism exists for collating and analysing these data comprehensively.

Health promotion and community engagement indicators are weakest, often collected through short-lived donor initiatives. As a result, annual national health sector reports overlook these areas, limiting accountability and resource allocation, despite being flagged as critical during recent post-emergency or after-action reviews.

Step 2. Cross-map existing indicators with EPHFs

Examples of key actions

- Identify and convene key stakeholders for the mapping exercise.
- Map currently monitored indicators across the nationally adopted or adapted EPHF framework. Use the EPHF subfunctions (3) or public health service areas for greater specificity.

The mapping can be done at national or subnational level. A sample mapping template is provided in the example scenario below, for adaptation as required.

Expected outputs

- Identification of which EPHFs are adequately measured and which remain underrepresented, in line with population health needs.
- Clearer understanding of overlaps, gaps and redundancies in monitoring.

Table 8 presents examples of country-level entry points for step 2, while Box 3 presents an example scenario for implementation of step 2 at country level.

Step 3. Identify and select additional indicators to close gaps

Examples of key actions

- Based on step 1 findings, define a complementary set of EPHF indicators to integrate into existing monitoring tools.
- Draw from EPHF indicators available in:
  - National sources, for example, health sector strategic plan, national action plan for health security, human resources for health policies, or primary health care monitoring frameworks *(29)*. These sources may contain relevant public health-focused indicators that had been previously approved but are yet to be implemented or rolled out at relevant levels.
  - International resources, for example, WHO's health system resilience indicators *(28)*, the primary health care measurement framework *(29)*, and the Sphere Handbook (for humanitarian contexts) *(23)*.
- Validate and adapt selected indicators in consultation with broad stakeholder input (including technical experts, programme managers and policymakers).

Selection of additional indicators and where necessary revision of existing indicators should be guided by population health needs and risks, feasibility of implementation, and an integrated, comprehensive approach to public health.

Expected output

- An approved list of indicators to be incorporated in existing monitoring and evaluation tools and processes (for example, national health information system, DHIS2, IDSR or annual health system performance reviews).

Table 9 presents examples of country-level entry points for step 3, while Box 4 presents an example scenario for implementation of step 3 at country level.

Box 4. Example scenario for implementation of step 3 (continued from step 2)

Based on the identified gaps, country X decides to improve routine monitoring of the health promotion and community engagement aspect of EPHFs.

The ministry of health's department of health and the national public health institute work together to convene stakeholders to review and select new indicators on community engagement and health promotion from various national and global sources, including reports on health sector reforms, past health sector plans and WHO resilience indicators.

Indicators are validated, adapted and integrated into DHIS2 for reporting at facility, district and national levels. Examples of the selected indicators are as follows:

- percentage of districts with an operational health promotion strategy;
- existence of a national budget line for health promotion;
- percentage of facilities reporting community participation in planning or feedback mechanisms;
- percentage of nomadic or cross-border populations receiving essential services, including health promotion and preventive services for priority health threats.

Table 9. Country-level entry point examples for step 3: Identify and select additional indicators to close gaps

Key country-level references to anchor and institutionalize EPHF indicators	Relevance
National policies and plans (health sector strategic plan, national action plan for health security, human resources for health policy, primary health care frameworks)	Sources of already approved but underutilized indicators that can fill EPHF gaps
Interministerial and intersectoral coordination platforms (universal health coverage, health security committees)	Ensure multisectoral ownership of indicators beyond ministry of health, especially for determinants of health
Ministry of health technical departments (health promotion, service delivery, primary health care, human resources for health, finance)	Custodians of programme-specific indicators; their buy-in ensures feasibility and uptake
Civil society and professional associations	Validate indicators against community needs, promote accountability and increase legitimacy
Regional or district health authorities	Test practicality of new indicators in front-line data collection and reporting

Step 4. Allocate resources and build capacity as part of integrated planning

Examples of key actions

- Assess the need for additional capacity-building in terms of human resources, infrastructure and funding.
- Update tools, software and infrastructure for EPHF data capture, analysis and use (for example, DHIS2 modules, mobile apps for community reporting).
- Ensure resources are available for disaggregation (sex, age, geography, vulnerability) to reflect health equity dimensions of EPHFs.
- Provide training and supervision for data producers and users at all levels, including front-line staff, monitoring and evaluation officers, policymakers and planners.

Sustainability should be a key consideration in all capacity-building efforts.

Expected outputs

- Availability and readiness of required resources and capacity to implement integrated measurement of EPHFs.

Table 10 presents examples of country-level entry points for step 4, while Box 5 presents an example scenario for implementation of step 4 at country level.

Box 5. Example scenario for implementation of step 4 (continued from step 3)

Based on the needs identified through stakeholder consultations:

- Country X ministry of health reviews and updates its health data monitoring and evaluation framework with a proportionate focus on EPHFs.
- The DHIS2 platform is updated, incorporating modules on health promotion and community engagement indicators focusing on public health.
- The country also updates its digital tool for collecting data at health facility level to include the newly adopted community engagement and health promotion indicators.
- Capacity-building workshops are held for health workers, community volunteers, and district monitoring and evaluation officers to ensure quality and consistency.

Table 10. Country-level entry point examples for step 4: Allocate resources and build capacity as part of integrated planning

Key country-level references to anchor and institutionalize EPHF indicators	Relevance
National budget and medium-term expenditure framework processes	Ensure sustainable financing for EPHF measurement, avoiding reliance on short-term donor funding
Donor and partner alignment platforms (Global Fund to Fight AIDS, Tuberculosis and Malaria; Gavi, the Vaccine Alliance; World Bank; bilateral partners)	Mobilize external resources and align donor monitoring and evaluation requirements with national EPHF indicators
Curriculum development and review of training institutions (for example, schools of public health, nursing, medicine)	Build long-term workforce capacity by embedding EPHF data competencies in curricula
District health management team functions	Operational level where data are collected and used – critical for sustainability and quality assurance

Step 5. Institutionalize measurement of EPHF indicators

Examples of key actions

- Integrate EPHF monitoring into existing mechanisms such as DHIS2, health sector performance reviews, monitoring and evaluation in health security and humanitarian contexts, and reporting on Sustainable Development Goals and universal health coverage.
- Avoid parallel systems to ensure that EPHFs become part of institutionalized data flows.
- Adapt measurement levels:
  - service-oriented EPHFs (for example, health promotion) tracked at facility and district levels;
  - enabling EPHFs (for example, governance, workforce, financing) monitored at national or subnational administrative level.

Expected outputs

- Delivery of EPHFs is monitored as part of an established, government-led health information system used in routine and emergency contexts, with proportionate emphasis on all domains.

Table 11 presents examples of country-level entry points for step 5, while Box 6 presents an example scenario for implementation of step 5 at country level.

Box 6. Example scenario for implementation of step 5 (continued from step 4)

- Country X includes and tracks selected health promotion and community engagement EPHFs as part of its routine health information (DHIS2).
- Data on the new EPHF indicators (along with the pre-existing EPHF indicators) are collected at health facility, subnational and national levels, as applicable, allowing policymakers to track not only clinical service outputs but also system capacities for public health.
- The health sector plan includes a clear budget line for public health, including funding for public health information management that covers all EPHFs.

Table 11. Country-level entry point examples for step 5: Institutionalize measurement of EPHF indicators

Key country-level references to anchor and institutionalize EPHF indicators	Relevance
National health information system policy or framework	Provides the legal and strategic mandate to integrate EPHFs into national reporting
Health sector performance reviews (joint health sector review, Sustainable Development Goal and universal health coverage reports)	High-visibility accountability platforms where EPHF performance can influence planning and funding
IHR monitoring tools (Joint External Evaluation, SPAR, after-action review)	Global compliance mechanisms that can be leveraged to institutionalize EPHF monitoring
Parliamentary oversight committees	Ensure political accountability, approval of budgets, and legitimacy of EPHF integration
Subnational health systems (provincial and district reporting mechanisms)	Critical for service-level EPHFs (for example, health promotion, surveillance) that require local-level tracking

Step 6. Analyse, report and use EPHF data comprehensively and inform national policy and planning

Examples of key actions

- Collate and analyse EPHF indicators across platforms to produce an integrated and comprehensive national and subnational (as applicable) picture of public health capacities.
- Present results in scorecards or dashboards that highlight performance by EPHF domain, including preventive, promotive, protective and public health emergency management.
- Use evidence to inform:
  - health sector planning and resource allocation for routine and emergency contexts;
  - multisectoral engagement (for example, education, agriculture, environment);
  - public accountability (annual reports, parliamentary briefings, civil society dissemination).

Expected outputs

- Readily available and up-to-date information on capacities and implementation of EPHFs, including a focus on all public health services domains.
- Evidence-based health sector and multisectoral policies, plans, resource allocation, accountability mechanisms and service packages that address public health capacity gaps and population health needs and risks.

Table 12 presents examples of country-level entry points for step 6, while Box 7 presents an example scenario for implementation of step 6 at country level.

Box 7. Example scenario for implementation of step 6 (continued from step 5)

- Country X's national health observatory develops an EPHF dashboard showing capacities in all EPHFs, including the new indicators on health promotion and community engagement. The dashboard is aligned with and draws on selected public health data from DHIS2, IDSR, and the IHR Monitoring and Evaluation Framework (SPAR, Joint External Evaluation).
- During public health emergencies, the EPHF dashboard is used in the emergency operations centre to guide decision-making and actions in response to and recovery from the emergency, leveraging and strengthening existing health system capacities for public health, for example in surveillance and community engagement.
- The utility of the dashboard is regularly reviewed and aligned with priority areas of improvement identified through national and international public health assessments and reviews. For example, regional and district health authorities are supported to adapt and manage the dashboard to guide public health actions at subnational levels.
- The dashboard informs the next health sector plan and budget, with allocation of new funds for health promotion, and reinforced interministerial collaboration with the education and communication sector. Revision of the county's package of essential health services incorporates clear roles of primary, secondary and tertiary care in promoting health and well-being and engaging the communities served. These changes lead to improved community participation and health promotion in routine health services and as part of public health emergency management, making the health system and population more resilient to stressors and shocks.

Table 12. Country-level entry point examples for step 6: Analyse, report and use EPHF data comprehensively and inform national policy and planning

Key country-level references to anchor and institutionalize EPHF indicators	**Relevance**
National health observatory, ministry of health dashboards	Central hub for data integration, analysis and dissemination of EPHF performance
Emergency operations centres	Ensure EPHF data are used in real time for preparedness, response and recovery
Ministry of health planning and budgeting cycles (annual health sector operational plans, budget hearings)	Turn EPHF evidence into resource allocation and programmatic decisions
Cross-sectoral councils (education, environment, agriculture)	Promote multisectoral action by linking health performance to determinants outside ministry of health
Civil society and media platforms	Translate data into public accountability and citizen engagement, driving demand for EPHF investments
Global and regional health bodies (WHO, International Association of National Public health Institutes, regional centres for disease control and prevention)	Enable benchmarking, peer learning and cross-country accountability for EPHF implementation.

5.4.2 Country examples and checklist

Country-level case examples are summarized in Table 13 with reference to the six steps discussed above. These cases show that while terminology may differ across contexts, countries are addressing measurement of EPHFs by embedding them into health information systems, governance structures, financing mechanisms and accountability processes. While most of the country examples do not match all the five steps, or prioritize all public health services domains, they collectively showcase the feasibility and potential benefits of integrating measurement of EPHFs in health information systems.

Table 13. Country-level case examples

Country and initiative	**What was done**	**Why it is notable**	**Steps matched (across the six-step process)**
1. Sierra Leone Integration of risk communication and community engagement into IHR monitoring and evaluation *(30)*	Post-Ebola and during COVID-19, Sierra Leone integrated risk communication and community engagement into its SPAR, Joint External Evaluation and after-action reviews, tracking indicators such as community trust, feedback systems and rumour tracking	Health promotion and community engagement were monitored as part of health security monitoring and evaluation, strengthening links between emergency preparedness and promotive EPHFs	Step 1: Secured political and leadership commitment Step 2: Mapped risk communication and community engagement elements into IHR core capacities Step 3: Introduced indicators on community engagement and trust Step 4: Built capacity for social listening and feedback analysis Step 5: Institutionalized risk communication and community engagement monitoring in national health security framework Step 6: Data used to adjust COVID-19 response strategies and build longer-term trust
2. China High-quality development of the disease control and prevention system *(2023–2030) (31)*	China issued national guidelines that strengthened the measurement backbone of its disease control and prevention system: modernizing real-time surveillance, improving early warning systems, integrating community engagement data, clarifying indicator responsibilities across disease control and prevention levels, and upgrading digital systems to ensure unified data reporting and analysis	China's reforms explicitly link public health functions (surveillance, workforce capabilities and multisectoral collaboration) to measurable performance indicators. It demonstrates how a national reform agenda can embed EPHF-oriented measurement into routine disease control and prevention governance at scale	Step 1: High-level political commitment was secured Step 2: Mapped gaps in existing indicator systems Step 3: Added new indicators (surveillance quality, rapid response, community engagement) Step 4: Built capacity through digital upgrades and workforce strengthening Step 5: Institutionalized measurement improvements through centre for disease control and prevention, restructuring and standardizing reporting Step 6: Used improved data for preparedness and system improvement

Table 13 (continued). Country-level case examples

Country and initiative	What was done	Why it is notable	Steps matched (across the six-step process)
3. Pan American Health Organization (PAHO) Renewal and mapping of essential public health functions *(32)*	In 2020–2021, PAHO led a regional process to update the EPHFs. Countries (including Chile, Colombia, Dominican Republic and Guatemala) conducted self-assessments to map existing indicators, identify gaps, and reorganize public health functions	A large-scale regional effort was made to embed EPHFs into health systems across multiple countries. Explicit links were made between EPHFs, health security and resilience post-COVID-19. Gaps were highlighted in health promotion, governance and financing	Step 1: Secured political and leadership commitment Step 2: Mapped national frameworks and indicators against EPHFs Step 3: Identified underrepresented functions (such as promotion, intersectoral action) Step 4: Regional training built technical and governance capacity Step 5: Institutionalized updated EPHFs as regional reference Step 6: Used mapping results in sector reviews, pandemic planning and resource allocation
4. Uganda IDSR and community event-based surveillance *(33)*	Expanded IDSR to include community event-based reporting, linked to DHIS2, covering outbreak-prone diseases and community health threats	Demonstrates how IDSR (traditionally epidemic focused) can capture broader public health functions, such as community engagement and early warning	Step 1: Secured political and leadership commitment Step 2: Mapped IDSR indicators to EPHFs Step 3: Added community engagement indicators Step 4: Institutionalized within DHIS2 Step 5: Used in real-time outbreak management
5. Ireland WHO EPHF review *(34)*	Reviewed existing monitoring and evaluation mechanisms as part of the EPHF strategic assessment across policies, structures, intersectoral coordination, monitoring and evaluation and services	First structured application of the EPHF in Ireland to inform public health reforms	Step 1: Secured political commitment to commission the EPHFs review to inform reforms Step 2: Mapped health sector monitoring and evaluation tools across the EPHFs Step 3: Identified strengths and gaps in current monitoring and evaluation coverage of EPHFs
6. Thailand Thai Health Promotion Foundation *(35)*	Earmarked 2% alcohol and tobacco levy funds a national health promotion agency with measurable programmes and indicator tracking	Demonstrates sustainable health promotion measurement and financing beyond donor cycles	Step 1: Secured political and leadership commitment Step 3: Introduced new promotion indicators not in routine health information system Step 4: Secured earmarked resources through legislation Step 5: Health promotion monitoring institutionalized via Thai Health mandate Step 6: Reports used for accountability to Parliament and public

Table 13 (continued). Country-level case examples

Country and initiative	What was done	Why it is notable	Steps matched (across the six-step process)
7. Bangladesh Humanitarian needs assessments (Rohingya response) (36)	Integrated health promotion, prevention and service access indicators into humanitarian health assessments	Shows that humanitarian monitoring and evaluation can track promotive and preventive functions, not only outbreak response	Step 1: Secured political and leadership commitment Step 3: Added EPHF indicators (promotion, access) Step 4: Built capacity for humanitarian data collection Step 6: Results used in health and intersectoral planning
8. Burkina Faso Nutrition indicators in DHIS2 (37)	Integrated new nutrition indicators into DHIS2; created dashboards; decentralized reporting and used routine data for performance review and decision-making	Demonstrates how a neglected area (nutrition, health promotion) can be mainstreamed into the routine health information system	Step 1: Secured political and leadership commitment Step 3: Gaps in nutrition identified – new indicators selected Step 4: Capacity built at district level with decentralized entry and data validation Step 5: Nutrition data institutionalized in national health management information system (not a donor silo) Step 6: Dashboards enabled evidence use during reviews
9. New Zealand Health system indicators framework and dashboard (38)	Indicators were co-designed with Māori representatives, clinicians, and health sector leaders; public dashboard integrated with New Zealand health survey data for small-area analysis	Example of equity-driven, participatory indicator design ensuring cultural relevance with strong public accountability	Step 1: Secured political and leadership commitment Step 4: strengthened data collection, especially for Māori and Pacific populations Step 5: Embedded indicators into Ministry of Health performance framework Step 6: Dashboard informs national and regional dialogue and enables public accountability
10. Brazil e-SUS AB and Previne Brasil (39)	Built nationwide primary health care information system (e-SUS AB); Previne Brasil links primary health care indicators to performance-based financing	Demonstrates how measurement and financing incentives can drive improvements in preventive and promotive services, not just curative care	Step 1: Secured political and leadership commitment Step 4: Digital primary health care infrastructure and training rolled out nationally Step 5: Indicators legally embedded into primary health care contracts Step 6: Performance results inform funding allocations and reforms
11. England (United Kingdom of Great Britain and Northern Ireland) Public Health Outcomes Framework (40)	Developed around 150 public health indicators, disaggregated by deprivation; published in dashboards	An example of embedding public health monitoring into national accountability systems	Step 1: Secured political leadership and commitment Step 5: Indicators are legally embedded in the Public Health Outcomes Framework and monitored by government Step 6: Results are routinely published for policy-makers, parliament and the public

5.5 Checklist for country-level use

This tool is designed to assist national and subnational health authorities and partners in reviewing how effectively the EPHFs are integrated into their monitoring and evaluation processes. It provides a set of key high-level questions with Yes/No prompts, followed by suggested actions, aligned with the above six-step process for strengthening EPHF measurement (Table 14).

Table 14. Checklist for integrating EPHFs into country-level monitoring and evaluation processes

Areas, guiding questions	Responses and actions
1. Is there a policy, strategy or plan to adopt and implement the EPHFs comprehensively, including application in monitoring and evaluation?	**Yes:** Engage stakeholders to sustain awareness and application of EPHFs in planning, service delivery and monitoring and evaluation **No:** Demonstrate leadership commitment to adopt and apply EPHFs in policymaking, planning, service delivery, and monitoring and evaluation *(step 1)*
2. Has there been a recent review o current health monitoring tools and approaches with regard to EPHF coverage and integration?	**Yes:** Ensure findings from the review are used to improve EPHF monitoring **No:** Map currently monitored indicators to EPHFs to identify overlaps, strengths and gaps *(step 2)*
3. Are the EPHF indicators systematically and comprehensively incorporated in current monitoring tools?	**Yes:** Confirm that existing indicators cover all EPHF domains **No:** Select and integrate additional EPHF indicators into existing monitoring frameworks (DHIS2, national health information system, IDSR, sector performance reviews) *(step 3)*
4. Does the health system have the capacity and resources to monitor and utilize EPHF data?	**Yes:** Confirm that monitoring spans all EPHF domains (service, workforce, governance, financing, health promotion and surveillance) **No:** Build capacity and allocate resources (human resources, tools, financing) for EPHF data collection, analysis and use *(step 4)*
5. Are EPHF indicators monitored as part of the national health information system, including monitoring and evaluation for health security?	**Yes:** Continue integrating EPHFs in national health information system and DHIS2 and align with IHR, Joint External Evaluation and SPAR monitoring **No:** Institutionalize EPHF monitoring into national health information system and avoid parallel systems *(step 5)*
6. Are EPHF indicators analysed and reported comprehensively, and are the data utilized to inform public health decisions and actions?	**Yes:** Sustain comprehensive analysis, reporting and data use; review and update as needed **No:** Establish systems for integrated analysis, reporting and use of EPHF data in decision-making *(step 6)*

6. Conclusion

The EPHFs represent an effective approach to prevent disease, promote and protect health and well-being, and address broad determinants of health in support of attaining universal health coverage, health security and healthier populations. Measurement of EPHFs enables understanding of current capacities and supports the targeting of investments to build resilient, equitable and accountable health systems. Existing health system and health security measurement tools provide significant valuable information on public health capacities across health promotion, health protection, disease prevention and early detection, public health emergency management, and community engagement and social participation functions that are context appropriate.

This technical report details the existing coverage of EPHFs through a comprehensive review of existing health system and health security measurement tools, including broad coverage of public health emergency management and health protection capacities across routine, voluntary and emergency tools. It highlights gaps requiring additional measurement, particularly in health promotion, disease prevention and integration of community engagement across routine public health delivery. It also recommends actions and tools, with case examples for country-level application, highlighting that institutionalizing EPHFs in monitoring and evaluation is not only about better data – it is also about ensuring that all public health functions are recognized, measured and used to inform decisions that protect and improve population health.

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Annex 1. Cross-mapping of 12 EPHFs and selected health system and health security assessment tools

Note: Partial levels of measurement are shaded yellow, while indicative levels of measurement are shaded green. Grey shading denotes where indicators in the tools do not capture the given public health subfunction.

HHFA = Harmonized Health Facility Assessment; IDSR = Integrated Disease Surveillance and Response: Pulse = pulse survey on continuity of essential health services during the COVID-19 pandemic; SPAR = States Parties self-assessment annual reporting; Sphere = Sphere Handbook; SPRP M&E = COVID-19 Strategic Preparedness and Response Plan: Global Monitoring and Evaluation Framework.

		Health System Assessment tool						
EPHF	**Subfunction**	**DHIS2**	**IDSR**	**SPAR**	**HHFA**	**Pulse**	**SPRP M&E**	**Sphere**
EPHF1	Subfunction 1.1: Planning for public health monitoring and surveillance	1.1	1.1	1.1			1.1	1.1
	Subfunction 1.2: Routine and systematic collection of public health data	1.2	1.2	1.2	1.2	1.2	1.2	1.2
	Subfunction 1.3: Analysing and interpreting available public health data		1.3	1.3	1.3		1.3	
	Subfunction 1.4: Communicating public health data, information and evidence with key stakeholders, including communities		1.4	1.4			1.4	
EPHF2	Subfunction 2.1: Monitoring and analysing available public health information to identify and anticipate potential and priority public health risks, including public health emergency scenarios	2.1	2.1	2.1			2.1	2.1
	Subfunction 2.2: Planning and developing capacity for public health emergency preparedness and response as part of routine health system functioning in collaboration with other sectors, including development of a national health emergency response operations plan		2.2	2.2	2.2	2.2	2.2	2.2
	Subfunction 2.3: Carrying out and coordinating effective and timely public health emergency response activities while supporting the continuity of essential functions and services		2.3	2.3	2.3	2.3	2.3	2.3
	Subfunction 2.4: Planning and implementing recovery from public health emergencies with an integrated health system strengthening approach					2.4	2.4	
	Subfunction 2.5: Engaging with affected communities and stakeholders in the public and private sectors and health and allied sectors as part of whole-of-government and whole-of-society approaches to public health emergency management		2.5	2.5			2.5	

		Health System Assessment tool						
EPHF	**Subfunction**	**DHIS2**	**IDSR**	**SPAR**	**HHFA**	**Pulse**	**SPRP M&E**	**Sphere**
EPHF3	Subfunction 3.1: Advocating public health-oriented planning, policies and strategies					3.1		
	Subfunction 3.2: Strengthening institutional public health structures for the coordination, integration and delivery of public health functions and services in the health and other sectors			3.2		3.2		
	Subfunction 3.3: Developing, monitoring and evaluating public health regulations and laws that act as formal, regulatory, institutional frameworks for public health governance, functions and services					3.3		
	Subfunction 3.4: Maintaining and applying public health ethics and values in governance							
EPHF4	Subfunction 4.1: Conducting evidenced-based health system planning and prioritization for managing population health needs, including alignment of national strategies, policies and plans for public health				4.1	4.1		
	Subfunction 4.2: Promoting integrated cross-sectoral prioritization and planning for public health with intersectoral accountability mechanisms and a Health in All Policies approach to manage population health needs			4.2				
	Subfunction 4.3: Promoting sustainable and integrated financing for public health by improving the generation, allocation and utilization of public and pooled funds to strengthen health system foundational capacities in all contexts			4.3	4.3	4.3		4.3
	Subfunction 4.4: Planning and developing appropriate infrastructure for meeting population health needs, including key services in health facilities (water, sanitation, waste, energy)			4.4	4.4	4.4	4.4	4.4
	Subfunction 4.5: Monitoring and assessment of policies and plans, financing of health systems, and multisectoral efforts for health that improve public health, promote equity and inclusion, and strengthen resilience			4.5	4.5	4.5		4.5
EPHF5	Subfunction 5.1: Developing, implementing, monitoring and evaluating regulatory and enforcement frameworks, including compliance with international legislation, and mechanisms for the protection of specified populations (for example, workers, patients, consumers) and the general public from health hazards			5.1	5.1			
	Subfunction 5.2: Conducting risk assessments, risk communication and other risk management actions needed for all manner of health hazards			5.2	5.2	5.2	5.2	5.2
	Subfunction 5.3: Monitoring, preventing, mitigating and controlling confirmed and potential health hazards	5.3	5.3	5.3	5.3	5.3	5.3	5.3
EPHF6	Subfunction 6.1: Designing, implementing, monitoring and evaluating interventions, programs, services and platforms for primary, secondary and tertiary prevention, including consideration of equity	6.1		6.1	6.1			6.1
	Subfunction 6.2: Integrating consideration of prevention and early detection into service delivery platform design or redesign			6.2	6.2		6.2	6.2
	Subfunction 6.3: Working with partners to support the development, implementation, and monitoring of legislation, policies and program activities aimed at reducing exposure to risk factors and promoting factors that prevent disease			6.3		6.3	6.3	6.3

		Health System Assessment tool						
EPHF	**Subfunction**	**DHIS2**	**IDSR**	**SPAR**	**HHFA**	**Pulse**	**SPRP M&E**	**Sphere**
EPHF7	Subfunction 7.1: Designing, implementing and evaluating specific interventions or programs to promote health, including changes in behaviour, lifestyle, practices, and the environmental and social conditions that promote health and reduce health inequities	7.1			7.1			
	Subfunction 7.2: Taking and supporting action, with partners, to address wider determinants of both communicable and noncommunicable diseases through a whole-of-government, whole-of-society approach, including increasing individual and community participation in health impacting decisions			7.2				
	Subfunction 7.3: Advocating, developing and monitoring legislation and policies aimed at promoting health and healthy behaviours and reducing inequities	7.3		7.3				
	Subfunction 7.4: Undertaking evidence-based advocacy and health communication to promote healthy behaviours and socioecological environments and build community trust							
EPHF8	Subfunction 8.1: Promoting participatory decision-making and planning for health and the promotion of societal change that enhances, promotes and protects health and well-being			8.1	8.1			
	Subfunction 8.2: Building community capacity for participating in public health planning, interventions, services, and preparedness and response measures		8.2	8.2	8.2	8.2		
	Subfunction 8.3: Monitoring and evaluation of community engagement in public health planning, interventions, services, and preparedness and response measures to promote equity and inclusion		8.3	8.3	8.3			
	Subfunction 8.4: Mobilizing and collaborating with communities and civil society groups in health services, interventions and programs as part of a whole-of-society approach		8.4	8.4		8.4		
	Subfunction 8.5: Engaging communities in health preparedness, readiness, response and recovery		8.5	8.5		8.5		
EPHF9	Subfunction 9.1: Undertaking planning and regular monitoring and evaluation of the public health workforce in relation to density, distribution and skills mix required to meet population health needs	9.1		9.1	9.1			9.1
	Subfunction 9.2: Assessing and developing the education and training of the public health workforce, encompassing the full spectrum of public health competencies (for example, technical, strategic and leadership skills), including development of essential competencies for intersectoral work for health and for emergency response			9.2	9.2	9.2		
	Subfunction 9.3: Promoting the sustainability of the public health workforce by developing appropriate career pathways and assessing and creating safe and dignified working conditions			9.3	9.3			

		Health System Assessment tool						
EPHF	**Subfunction**	**DHIS2**	**IDSR**	**SPAR**	**HHFA**	**Pulse**	**SPRP M&E**	**Sphere**
EPHF10	Subfunction 10.1: Assessing and improving the quality and appropriateness of health services and social care services as delivered to meet population health needs	10.1		10.1	10.1	10.1		10.1
	Subfunction 10.2: Assessing and promoting equity in the provision of and access to health and social care services	10.2			10.2			10.2
	Subfunction 10.3: Aligning the planning and delivery of health services and social care services with population health needs and priority risks			10.3	10.3	10.3		
EPHF11	Subfunction 11.1: Strengthening and broadening the capacity to conduct and promote research in order to enhance the knowledge base and inform evidence-based policy, planning, legislation, financing and service delivery at all levels and in all contexts	11.1						
	Subfunction 11.2: Supporting knowledge development and implementation, including the translation of public health research into decision-making based on the best available evidence and practices for addressing population health needs							
EPHF11	Subfunction 11.3: Promoting the inclusion and prioritization of public health operational research within broader research agendas							
	Subfunction 11.4: Promoting and maintaining ethical standards in public health research that promote a human rights-based approach to health							
EPHF12	Subfunction 12.1: Developing and implementing policies, laws, regulations and interventions that promote the development of and equitable access to essential medicines and other medical products and health technologies in both national and international contexts			12.1	12.1			12.1
	Subfunction 12.2: Developing and implementing evidence-based standards, laws, regulations, policies and interventions that ensure the safety, affordability and efficacy of essential medicines and other medical products and health technologies	12.2		12.2	12.2			
	Subfunction 12.3: Working with partners to manage the inclusion of evidence-based essential medicines and other medical products, health technologies and non-pharmacological interventions into clinical and public health practices	12.3		12.3	12.3	12.3		
	Subfunction 12.4: Managing supply chains for essential medicines and other medical products and health technologies in support of their rational use and equitable access in both national and international contexts, including stockpiling and prepositioning essential medicines, equipment and supplies	12.4		12.4	12.4	12.4		12.4
	Subfunction 12.5: Monitoring and assessing the safety, effectiveness, efficacy, and utilization of, and access to, essential medicines and other medical and surgical products, health technologies and non-pharmacological interventions, in clinical and public health settings				12.5	12.5		12.5

Annex 2. Technical brief to inform policies and planning

Strengthening integrated measurement of essential public health functions-aligning health systems and health security monitoring for resilient, people-centred public health

The report "Measuring integrated delivery of essential public health functions: a review of health systems and health security monitoring tools" *(1)* funded through the International Health Grants Program of the Public Health Agency of Canada, serves as the foundation for this Brief.

Preview

Essential Public Health Functions (EPHFs) are increasingly recognized as foundational to resilient, equitable, and people-centered health systems. However, current approaches to monitoring public health capacity remain limited and fragmented across health systems and health security frameworks. This technical brief draws on a review of widely used routine, voluntary and emergency-driven monitoring tools and existing WHO normative guidance, to highlight why and how countries can strengthen the integrated measurement of EPHFs in support of universal health coverage, health security, and population health.

Key messages

- EPHFs are central to resilient health systems but are not measured in an integrated way.
- Existing monitoring tools capture elements of EPHFs but remain fragmented and siloed.
- Gaps are most evident in health promotion, prevention beyond clinical care, community engagement, and recovery aspect of emergency management.
- Integrated measurement improves visibility, accountability, and investment in public health.
- Countries can strengthen EPHF measurement by better aligning data from existing tools while adopting plans to incorporate more focused indicators for EPHFs and public health services.

Background and challenge

Recent public health shocks, including the COVID-19 pandemic, exposed persistent weaknesses in public health systems and core health system functions, including routine monitoring and oversight *(2, 3)*. Although countries routinely generate large volumes of data through health systems and health security monitoring mechanisms, these data are rarely organized to reflect the comprehensive and integrated delivery of Essential Public Health Functions (EPHFs) across health and allied sectors.

As a result, public health capacities are often assessed in isolation, with monitoring frameworks prioritizing emergency response over prevention, health promotion, and recovery. Contributions from sectors beyond health also remain poorly visible, limiting accountability and evidence-informed decision-making *(4, 5)*.

Member States and WHO have committed to strengthening EPHFs through World Health Assembly Resolution WHA69.1 and the Fourteenth General Programme of Work (GPW14, 2025–2028) *(6, 7)*. Strengthening the integrated measurement of EPHFs is therefore essential to translate these commitments into effective policy-making, planning, financing, and service delivery.

What the evidence shows

Based on a review of seven widely used monitoring tools assessed through the EPHF lens, existing health systems and health security measurements capture important aspects of public health capacity but do so unevenly and with limited integration across functions *(1)*. The review covered routine, voluntary, and ad hoc tools, including District Health Information Software 2 (DHIS2), Integrated Disease Surveillance and Response (IDSR), IHR States Parties Self-Assessment Annual Reporting (SPAR), Harmonized Health Facility Assessment (HHFA), WHO Pulse Surveys, as well as recommended indicator sets from the COVID-19 Strategic Preparedness and Response Plan and the Sphere Handbook.

Areas of relative emphasis

- Public health emergency management is the most consistently measured function, particularly through health security-related tools.
- Health protection and surveillance are well covered for infectious and selected environmental hazards.
- Community engagement is increasingly measured in emergency and health security contexts, notably through risk communication capacities.

Persistent gaps

- Health promotion, especially evidence-based advocacy and health communication, is weakly measured.
- Disease prevention beyond clinical services, including action on social and commercial determinants of health, is insufficiently captured.
- Community and multisectoral engagement in routine settings remains poorly measured outside emergencies.
- Recovery and resilience following public health emergencies receive limited systematic attention.
- Equity and inclusion are inconsistently embedded across monitoring frameworks.

No single monitoring tool captures the integrated delivery of EPHFs, reinforcing fragmentation across health agendas and accountability mechanisms.

Why integrated measurement matters

Measurement shapes priorities. When EPHFs are not measured together, prevention and health promotion are deprioritized, community trust and participation are overlooked, and investments prioritize response over baseline capacities for resilience.

Integrated measurement of EPHFs supports:

- Coherence between universal health coverage, health security, and population health and wellbeing agendas
- More efficient use of existing data and resources which are continuously constrained in many settings
- Stronger accountability for public health outcomes across health and allied sectors
- Health systems resilience covering preparedness against ongoing and future shocks.

Policy actions and operational guidance

Countries can strengthen EPHF measurement through coordinated actions at measurement and governance levels, building on existing WHO guidance and tools[1].

1. Strengthen measurement approaches

Countries, with the support of partners as applicable, should:

- Adapt current tools (e.g. DHIS2, HHFA, SPAR) to better capture health promotion, primary prevention, and community engagement in all contexts.
- Introduce targeted indicators for health communication, trust-building, and post-emergency recovery and resilience.
- Apply an all-hazards measurement approach - beyond infectious threats - reflecting the broad range of context-relevant public health risks and required services
- Expand metric development beyond the health sector, capturing health promotion and protection functions delivered through collaboration with allied sectors
- Embed equity across public health related measurement processes

A phased approach—building from indicative to more comprehensive measures—supports feasibility and sustainability.

2. Strengthen stewardship and coordination

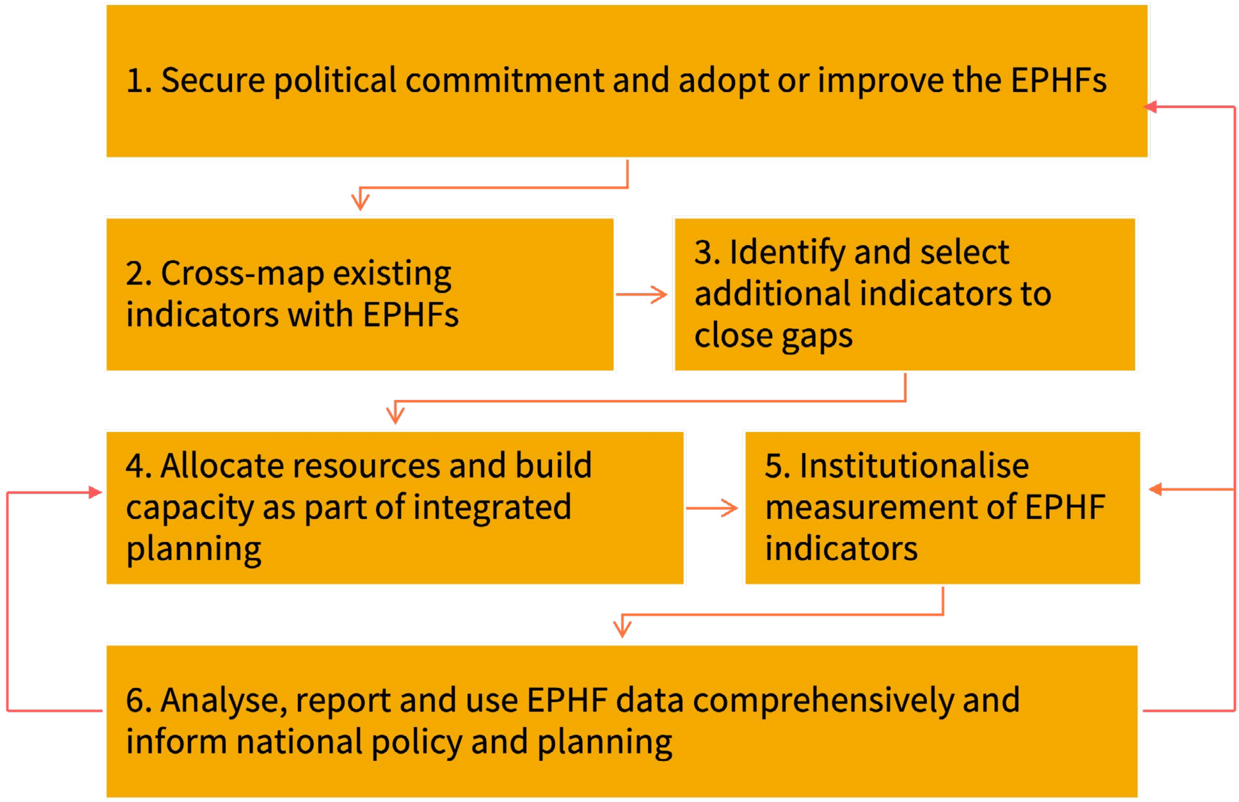
Countries, with the support of partners as applicable, should:

- Mandate responsible national authorities to regularly review national health monitoring frameworks through a comprehensive public health lens (EPHFs).
- Integrate EPHFs explicitly into health strategies, health security plans, and recovery frameworks as informed by population health risks and burden of diseases.
- Strengthen multisectoral coordination mechanisms for public health measurement.
- Invest in workforce capacity for integrated monitoring, analysis, and data use.
- Align partner and donor investments with integrated EPHF priorities informed by population health needs.
- Invest in and sustain community engagement in public health planning, actions and monitoring.

3. Implement the following interdependent steps to operationalize EPHFs integration in existing monitoring processes (Figure A2.1)

[1] WHO's guidance and tools on essential public health functions and health systems resilience can accessed through World Health Organization. Health Systems Resilience [website]. Geneva: World Health Organization; 2026 (https://www.who.int/teams/primary-health-care/health-systems-resilience, accessed 16 April 2026).

Figure A2.1. Key steps for integrating EPHFs in existing monitoring processe



Conclusion

Countries do not need new measurement systems. By re-orienting and integrating existing health systems and health security monitoring tools around Essential Public Health Functions, governments can strengthen public health foundations, improve resilience, and advance healthier populations in a sustainable and cost-effective manner.

Table A2.1. Checklist for integrating EPHFs into country level M&E processes

Areas, guiding questions	Responses and actions
1. Is there a policy, strategy or plan to adopt and implement the EPHFs comprehensively, including application in monitoring and evaluation?	**Yes:** Engage stakeholders to sustain awareness and application of EPHFs in planning, service delivery and monitoring and evaluation **No:** Demonstrate leadership commitment to adopt and apply EPHFs in policymaking, planning, service delivery, and monitoring and evaluation *(step 1)*
2. Has there been a recent review o current health monitoring tools and approaches with regard to EPHF coverage and integration?	**Yes:** Ensure findings from the review are used to improve EPHF monitoring **No:** Map currently monitored indicators to EPHFs to identify overlaps, strengths and gaps *(step 2)*
3. Are the EPHF indicators systematically and comprehensively incorporated in current monitoring tools?	**Yes:** Confirm that existing indicators cover all EPHF domains **No:** Select and integrate additional EPHF indicators into existing monitoring frameworks (DHIS2, national health information system, IDSR, sector performance reviews) *(step 3)*
4. Does the health system have the capacity and resources to monitor and utilize EPHF data?	**Yes:** Confirm that monitoring spans all EPHF domains (service, workforce, governance, financing, health promotion and surveillance) **No:** Build capacity and allocate resources (human resources, tools, financing) for EPHF data collection, analysis and use *(step 4)*
5. Are EPHF indicators monitored as part of the national health information system, including monitoring and evaluation for health security?	**Yes:** Continue integrating EPHFs in national health information system and DHIS2 and align with IHR, Joint External Evaluation and SPAR monitoring **No:** Institutionalize EPHF monitoring into national health information system and avoid parallel systems *(step 5)*
6. Are EPHF indicators analysed and reported comprehensively, and are the data utilized to inform public health decisions and actions?	**Yes:** Sustain comprehensive analysis, reporting and data use; review and update as needed **No:** Establish systems for integrated analysis, reporting and use of EPHF data in decision-making *(step 6)*

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